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Exploring a Dynamic Homotopy Technique to Enhance the Convergence of Classical Power Flow Iterative Solvers in Ill-Conditioned Power System Models

Alisson Lima-Silva  and Francisco Damasceno Freitas * 

Department of Electrical Engineering, University of Brasilia, Brasilia 70910-900, Brazil; alisson.lima@aluno.unb.br

* Correspondence: ffreitas@ene.unb.br

Abstract: This paper presents a dynamic homotopy technique that can be used to calculate a preliminary result for a power flow problem (PFP). This result can then be used as an initial estimate to efficiently solve the PFP using either the classical Newton-Raphson (NR) method or its fast decoupled version (FDXB) while still maintaining high accuracy. The preliminary stage for the dynamic homotopy problem is formulated and solved by employing integration techniques, where implicit and explicit schemes are studied. The dynamic problem assumes an initial condition that coincides with the initial estimate for a traditional iterative method such as NR. In this sense, the initial guess for the PFP is adequately set as a flat start, which is a starting for the case when this initialization is of difficult assignment for convergence. The static homotopy method requires a complete solution of a PFP per homotopy pathway point, while the dynamic homotopy is based on numerical integration methods. This approach can require only one LU factorization at each point of the pathway. Allocating these points properly helps avoid several PFP resolutions to build the pathway. The hybrid technique was evaluated for large-scale systems with poor conditioning, such as a 109,272-bus model and other test systems under stressed conditions. A scheme based on the implicit backward Euler scheme demonstrated the best performance among other numerical solvers studied. It provided reliable partial results for the dynamic homotopy problem, which proved to be suitable for achieving fast and highly accurate solutions using both the NR and FDXB solvers.

Keywords: dynamic homotopy; Newton-Raphson method; integration method; backward Euler scheme; fast decoupled Newton-Raphson; ill-conditioned system



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1. Introduction

The power flow computational tool is necessary to determine the operating point of an electrical grid. This nonlinear problem has been widely studied for a long time. Currently, it can involve large-scale, sometimes poorly conditioned systems, which can constantly change in some form depending on the grid conditions. More recent physical aspects that contribute to this change include the presence of new types of energy sources and the configuration of the grid to meet certain operational states. In this sense, research into robust computational techniques to solve the problem becomes attractive.

The main method for solving the power flow problem (PFP) is the iterative Newton-Raphson (NR) method and its variations, such as decoupled and fast decoupled techniques [1]. However, solving the resulting nonlinear problem can be challenging due to the difficulty in assigning the initial estimate. If the assignment is not properly defined, the problem might experience divergence or yield impractical results. Flat start initial estimation for the profile of electrical networks is a common strategy when precise knowledge of the system's operating point is difficult to obtain [1]. This is a more realistic situation nowadays due to changes caused by events such as the diversified load profile and the integration of intermittent renewable sources [2–5], among other factors [6,7]. In

certain cases, particularly in large networks, using flat start PFP initialization may cause the problem to diverge [8]. According to [9], when this numerical detail is identified, it does not necessarily mean the problem has no solution. i.e., if a solution exists, the problem is considered ill-conditioned. Therefore, finding the solution using traditional iterative techniques is challenging, that is, if the solution to the PFP cannot be obtained using a standard technique such as the classical Newton-Raphson method with a flat-start initial guess, the problem is considered ill-conditioned, even though a valid solution exists.

Investigations on the ill-conditioned topic applied to power system models have long been of interest [10]. However, in recent years numerous studies on the subject have been published [8,9,11–15]. Furthermore, there exist various methodologies proposing a way to solve the nonlinear problem embedded in the PFP [16].

A dynamic approach from the PFP power balance equations was investigated in [9] after the original problem was converted into a set of Ordinary Differential Equations (ODEs). In a first approach, the dynamic problem was solved using an explicit integration method [9]. Later on, the same problem was readdressed and resolved using an implicit method [8]. In these cases, the classical methods of Euler and Runge-Kutta were explored. Given the simplicity of these numerical integration techniques, other forms were investigated by other researchers [11,17]. The method developed in [11] considered the study of well- and ill-conditioned electrical systems models using a technique denominated Heun-King-Werner (HKW) method. The HKW method presented superior efficiency and convergence performance compared to conventional PFP solution methods. The classical Newton-Raphson method often struggles to converge when dealing with ill-conditioned systems, unless specific parameters and/or initializations are used. This dependency on certain conditions makes the convergence highly insecure for using the Newton-Raphson method or other alternatives.

Methods based on homotopy have also been recently developed, in contrast to the previously mentioned fictitious dynamic techniques [12,14]. The problem involves calculating a PFP solution along a homotopy pathway until its final point using the NR solver. Changes in the homotopy parameter along the pathway result in solving the problem, which is finally concluded when the homotopy parameter reaches a unitary value. Hence, how an algorithm is implemented along the homotopy path differs for each technique. For instance, in [14], the homotopy technique is applied in analogy with components of electric circuits to simulate the nonlinear equations of the power balance equations. The publication exploits a static homotopy approach to calculating the PFP solution along the homotopy pathway. In the study in [12], the static homotopy problem for the original PFP power balance equations is solved differently. As the homotopy parameter increases, the grid is transformed from a fictitious and decoupled bus power system to the original problem along the homotopy parameter changes. The PFP solution is obtained when the homotopy parameter reaches unity. In [18], the continuation homotopy technique is applied to trace curves and determine isolated PFP solutions. The work [19] uses homotopy techniques to solve the PFP using a technique in which the power balance equations of the power flow equations are embedded as a homotopy problem starting in an initial point until the solution. The security assessment problem is investigated in [20] by using the homotopy technique. The problem is associated to the robustness of the network operating point and its condition depends on the calculations of power flow solutions spaces's region. [20] uses the homotopy concept to find all solution boundaries, delimiting in regions for a system [21] proposed a numerical polynomial homotopy continuation method which is appropriate to determine all the solutions of a power flow. It is similar to the homotopy technique employed in [20], but requiring a large computational cost. Besides, both works were validated for small-scale test systems.

Some publications on homotopy techniques were also applied to the power flow problem in distribution systems [22,23]. Other applications are found to calculate Optimal Power Flow (OPF) [24,25] and for post-contingency actions as those verified in stressed network [26].

In the previous discussion of homotopy techniques, it was mentioned that solving the power balance equations is crucial at each point along the homotopy pathway. This procedure involves iterations to convergence at each point of the pathway and its convergence can be slow, demanding a large computational burden, mainly for large-scale systems. Or even presents divergence, especially when using directly the NR method. For this case, it is important to emphasize that each NR iteration requires an LU factorization of the Jacobian matrix, which is computationally intensive in PFP. So, computing the PFP solution using the static homotopy technique involves solving a series of nonlinear equation systems one after the other. Each new solution uses the previous result as an initial estimate. Reducing the number of points in the pathway can minimize the use of the solver, which in turn reduces the total number of iterations needed to obtain an effective solution for the PFP. An alternative is to relax the partial PFP results along the homotopy path. An investigation in this direction was proposed in [27], but instead of using a static homotopy-based technique, a dynamic homotopy approach [28] was studied. However, despite reducing the computational cost in combination with an NR solver to solve the same PFP in comparison with the traditional static technique, several aspects regarding the parameter-dependent method were defined empirically.

This paper presents a dynamic homotopy-based approach, together with classical iterative techniques, to compute the solution of the PFP. Based on this technique, in combination with the NR method, or using its simplified version called fast decoupled NR (FDXB), high-accuracy solutions for the PFP are obtained. The dynamic homotopy problem, as formulated in [27] and based on [28], was extended in this paper to investigate in-depth and implement explicit integration techniques, such as forward Euler (FE) and second-order Runge-Kutta (RK2), and an implicit integration technique based on the backward Euler (BE) scheme. The dynamic homotopy structure allows the original PFP to be initialized from a flat starting estimate. The problem formulation describes how a simple homotopy function using a fixed point structure was implemented. This procedure introduces a proportional term, which, as demonstrated in this proposed paper, is associated with a perturbation to remove or improve the numerical conditioning of the Jacobian matrix in the PFP (as suggested in [29]). The finding allows the solution of large-scale ill-conditioned problems demanding a reduced computational burden. The simulations show that the implicit method adequately provides an initial estimate for the NR and FDXB solvers. Furthermore, differently from the static homotopy-based problem, the formulation only requires one LU factorization for a dynamic homotopy pathway approximation point and allows for high integration steps, when the implicit BE scheme is used. To demonstrate the effectiveness of the results obtained in the investigations, simulations are conducted in poorly conditioned systems and networks under stressful operating conditions. The simulations are performed in large-scale systems, including a grid with more than 109k buses.

Some contributions are identified as follows:

- Presentation of a technique enabling the use of the NR and FDXB solvers, demonstrating their efficiency despite starting from a flat initial guess;
- Demonstration that the dynamic homotopy problem for the PFP can be transformed into a fixed-point function, where the proportional term multiplying this function is associated with the original Jacobian PF;
- Use of the result obtained from the dynamic homotopy directly to execute the native scripts for NR and FDXB in the MATPOWER software [30]; and
- Investigations of explicit and implicit integration techniques to solve the dynamic homotopy problem efficiently.

The paper is structured as follows: Section 2 was dedicated to succinctly presenting the fundamentals required for the basic definitions found in the PFP, including the presentation of some solvers. In Section 3, the homotopy problem is presented and then the dynamic homotopy problem is formulated. In this section, the problem associated with the fixed-point inclusion is described, and also a generic tutorial example is introduced. Tests

with several experiments and discussions of results are presented in Section 4. The main conclusions of the paper are described in Section 5.

2. Fundamentals of the Iterative Power Flow

A power system is typically described by power balance equations (PBEs), a set of nonlinear equations governing the system. Well-accepted modeling is presented as a set of equations representing the power flow entering (leaving) a bus, using a voltage polar representation. These equations can be generalized for a total of N_b buses by (1) for PQ- and PV-type, and (1) for only PV-type, where the load buses and generation buses are named as PQ- and PV-type, respectively.

$$0 = P_k^{sp} - V_k \sum_{m=1}^{N_b} V_m [G_{km} \cos(\theta_{km}) + B_{km} \sin(\theta_{km})], \quad (1)$$

$$0 = Q_k^{sp} - V_k \sum_{m=1}^{N_b} V_m [G_{km} \sin(\theta_{km}) - B_{km} \cos(\theta_{km})]. \quad (2)$$

The variable to be determined is the voltage phasor, at a generic bus k , given by $\bar{V}_k = V_k \angle \theta_k$. According to the phasor notation, V_k is the voltage magnitude, while θ is the phase angle. The power flow is represented by equations requiring the contribution of branches and shunt admittances. These admittances are organized in the form of an admittance matrix, $\mathbf{Y}_{bus}$, whose entries are given by $\bar{Y}_{km} = G_{km} + jB_{km}$. The net specified power injected into the bus k is given by P_k^{sp} to active power and Q_k^{sp} to reactive power. Both P_k^{sp} and Q_k^{sp} can assume voltage-dependent expressions [1] that are arranged into an exponential or a polynomial form, being the latter representation well-known as ZIP model. This work uses the representation from the MATPOWER framework, consisting of constant power and constant impedance, as available in the framework's databank.

In addition to the buses PQ and PV, the system was modeled with only one slack bus. Hence, the total number of buses in the system is given by $N_b = N_{PQ} + N_{PV} + 1$ and the total number of Equations (1) and (2) is given by $n = 2N_{PQ} + N_{PV}$. Therefore, to solve the problem, it is sufficient to find the roots of the power balance equations. This set of equations can be generically represented by $\mathbf{g}(\mathbf{x}) = \mathbf{0}$ and the solution can be given by $\mathbf{x} = [\theta^T \mathbf{V}^T]^T$ where $\mathbf{x} \in \mathbb{R}^n$ is the network state vector and $\mathbf{g}(\mathbf{x})$ is a map of the type $\mathbf{g}(\mathbf{x}) : \mathbb{R}^n \mapsto \mathbb{R}^n$, defining the mismatches (residues) of the PBEs.

2.1. Traditional Numerical Solvers

The PBEs can be solved by iterative numerical methods starting from an initial estimate $\mathbf{x}^{(0)}$ [31]. The iterative methods are characterized by increments and the objective is to find the root $\mathbf{x}_*$ of the equation $\mathbf{g}(\mathbf{x})$ by using an appropriate root-finder. The initial estimate $\mathbf{x}^{(0)}$ converge along the iteration to the root $\mathbf{x}_*$ when $\|\mathbf{g}(\mathbf{x}^{(i+1)})\|_\infty < \epsilon$, for $\epsilon > 0$ sufficiently small, a given tolerance. Usually, the number of iterations is limited, and the method is interrupted if a given threshold is exceeded.

Different methods proposed in the literature can solve the PBE [31]. Techniques such as Gauss-Seidel (GS) [31], and Newton-Raphson (NR) [1,32] are capable of numerically solving PBEs with different characteristics such as computation burden and number of iterations. The NR method stands out for presenting quadratic convergence when the initial voltage estimate is within the region of attraction. For this reason, it is widely used as a reference for performance analysis of numerical solvers.

2.1.1. Classical Newton-Raphson Method

The Newton-Raphson method is a classical technique used to solve the PBE [32]. It starts by using Taylor's series expansions to approximate nonlinear functions while avoiding higher-order terms [33]. From the initial estimate $\mathbf{x}^{(0)}$, the solution $\mathbf{x}^{(i)}$ is incremented

through the mismatches of the previous iteration to reach a more accurate solution $\mathbf{x}^{(i+1)}$. Knowing that the Jacobian matrix for $\mathbf{g}(\mathbf{x}) \in \mathbb{R}^n$ and $\mathbf{x} \in \mathbb{R}^n$ is $\mathbf{J}(\mathbf{x}^{(i)}) = \left. \frac{\partial \mathbf{g}(\mathbf{x})}{\partial \mathbf{x}} \right|_{\mathbf{x}^{(i)}}$, where

$$\mathbf{J}(\mathbf{x}^{(i)}) = \begin{bmatrix} \frac{\partial \mathbf{g}_1(\mathbf{x})}{\partial \mathbf{x}_1} & \frac{\partial \mathbf{g}_1(\mathbf{x})}{\partial \mathbf{x}_2} & \dots & \frac{\partial \mathbf{g}_1(\mathbf{x})}{\partial \mathbf{x}_n} \\ \frac{\partial \mathbf{g}_2(\mathbf{x})}{\partial \mathbf{x}_1} & \frac{\partial \mathbf{g}_2(\mathbf{x})}{\partial \mathbf{x}_2} & \dots & \frac{\partial \mathbf{g}_2(\mathbf{x})}{\partial \mathbf{x}_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial \mathbf{g}_n(\mathbf{x})}{\partial \mathbf{x}_1} & \frac{\partial \mathbf{g}_n(\mathbf{x})}{\partial \mathbf{x}_2} & \dots & \frac{\partial \mathbf{g}_n(\mathbf{x})}{\partial \mathbf{x}_n} \end{bmatrix}, \quad (3)$$

and the deviation $\Delta \mathbf{x}^{(i)} = -[\mathbf{J}(\mathbf{x}^{(i)})]^{-1} \mathbf{g}(\mathbf{x}^{(i)})$ refines the next estimate from the initial estimate where $\mathbf{x}^{(i+1)} = \mathbf{x}^{(i)} + \Delta \mathbf{x}^{(i)}$. If the Jacobian matrix is nonsingular, the iterative solution of the NR scheme can be summarized as

$$\mathbf{x}^{(i+1)} = \mathbf{x}^{(i)} - [\mathbf{J}(\mathbf{x}^{(i)})]^{-1} \mathbf{g}(\mathbf{x}^{(i)}). \quad (4)$$

The Fast Decoupled Power Flow (FDPF), initially proposed by Stott & Alsac [34], presents fast convergence compared to the full NR solver. with some approximations regarding impedance as $\mathbf{R}_{km} \ll \mathbf{X}_{km}$, the line conductance $\mathbf{G}$ is neglected of $\mathbf{Y}_{bus}$ and only susceptance elements will compose it, represented by $\mathbf{Y}_{bus} = j\mathbf{B}$, and the angular differences are considered near to zero, $\theta_{km} = \theta_k - \theta_m \simeq 0$. Considering the following approximations $\sin(\theta_{km}) \simeq \theta_{km}$ and $\cos(\theta_{km}) \simeq 1$, we have

$$\mathbf{P}_k = \sum_{\substack{m=1 \\ m \neq k}}^{N_b} V_k V_m \mathbf{B}_{km} (\theta_k - \theta_m), \quad (5)$$

$$\mathbf{Q}_k = - \sum_{m=1}^{N_b} V_k V_m \mathbf{B}_{km}. \quad (6)$$

In this way, it is possible to note that active power is associated with the angular phase shift ($\mathbf{P}\theta$), and reactive power is associated with the voltage magnitude variation ($\mathbf{QV}$). For convenience, the matrix $-\mathbf{B}$ in the active and reactive power equation is called $[\mathbf{B}']$ and $[\mathbf{B}'']$, respectively.

Summarizing the Equations (5) and (6) present the iterative equations can have the iterative process individualized, helping with the computational cost and can be organized as

$$\left[\frac{\Delta \mathbf{P}}{V} \right] = [\mathbf{B}'] [\Delta \theta], \quad (7)$$

$$\left[\frac{\Delta \mathbf{Q}}{V} \right] = [\mathbf{B}''] [\Delta V]. \quad (8)$$

Considering the FDXB approach [35], the resistance is ignored only in matrix $\mathbf{B}'$ formation.

Nonstandard iterative techniques can be found in the literature, for example, those referred to as Newton's Continuous (NC) type [8,9]. This methodology was inspired by the discrete iterative response in the process resolution carried out for a solution of an Ordinary Differential Equation (ODE) with a continuous parameter t . As a result, the ODE is represented by $\dot{\mathbf{x}}(t) = -[\mathbf{J}(\mathbf{x}(t))]^{-1} \mathbf{g}(\mathbf{x}(t)) = \Phi(\mathbf{x}(t))$. Now, it can be solved by a traditional numerical integration method considering a time-step Δt .

Some traditional integration techniques are briefly described in the sequence, considering the embedding of the PFP formulation.

2.1.2. Solution by Using a Standard Integration Method

The well-known numerical integration forward Euler (FE) method was adapted as a tool to solve the PFP in [9]. The static problem was converted to a “dynamic form” assuming a time-step Δt . Note that the traditional NR method is identified if $\Delta t = 1$. An iteration scheme of the FE, for a starting-time $\mathbf{x}^{(0)}$ is given by

$$\mathbf{x}^{(i+1)} = \mathbf{x}^{(i)} - \Delta t \left[\mathbf{J}(\mathbf{x}^{(i)}) \right]^{-1} \mathbf{g}(\mathbf{x}^{(i)}), \quad i = 0, 1, \dots \quad (9)$$

In (9), the initial estimate $\mathbf{x}^{(0)}$ for the static NR is also the initial condition for the ODE.

Since the FE is an explicit integration method, its performance when dealing with ill-conditioned systems in general has an aspect unfavorable. Therefore, an implicit scheme is more adequate. Exploiting this characteristic, a backward Euler (BE) type solver was proposed in [8]. The standard scheme for the BE is

$$\mathbf{x}^{(i+1)} = \mathbf{x}^{(i)} - \Delta t \left[\mathbf{J}(\mathbf{x}^{(i+1)}) \right]^{-1} \mathbf{g}(\mathbf{x}^{(i+1)}), \quad i = 0, 1, \dots \quad (10)$$

In (10) the upgrade of the state variable, $\mathbf{x}^{(i+1)}$, depends on the resolution of a nonlinear equation $\left[\mathbf{J}(\mathbf{x}^{(i+1)}) \right] \left[\mathbf{x}^{(i+1)} - \mathbf{x}^{(i)} \right] + \Delta t \mathbf{g}(\mathbf{x}^{(i+1)}) = 0$, where $\mathbf{x}^{(i+1)}$ is unknown and $\mathbf{x}^{(i)}$ is given. When dealing with large-scale problems the resolution of the nonlinear algebraic equation is not straightforward. In [8], it was presented some schemes to obtain a solution of this equation, including simplification aspects.

Techniques considering an aspect of predictor-corrector scheme are also found in the literature. It is important to consider that as the scheme becomes more complex, the computational workload to determine a single discrete value for the ODE can increase. This is due to the fact that each evaluation of the function $\Phi(\mathbf{x}(t))$ requires the inversion of a Jacobian matrix.

Numerical schemes based on Runge-Kutta method families were also published in [9]. Similarly to the FE method, the Runge-Kutta formulas are explicit integration methods. However, depending on its adopted order, it will require more evaluations of the function $\Phi(\mathbf{x}(t))$. The solver requires additional computation expressed by constants K_n for the same iterate. The second-order Runge-Kutta (RK2) iterative formula is given by

$$\mathbf{x}^{(i+1)} = \mathbf{x}^{(i)} + \frac{\Delta t(K_1 + K_2)}{2}, \quad (11)$$

where the constants are calculated as $K_1 = \mathbf{g}(\mathbf{x}^{(i)}, t^{(i)})$, and $K_2 = \mathbf{g}(\mathbf{x}^{(i)} + \Delta t K_1, t^{(i)} + \Delta t)$; $t^{(i)}$ is a fictitious time related to the iterations, generally not used explicitly in the formulas.

2.2. The Limitation of Traditional Methods

Traditional methods can have difficulty with convergence in practical power systems, especially in large-scale systems. For instance, techniques like Gauss-Seidel (GS), used to initialize the NR solver in some production-grade software, may experience slow convergence or be impractical in complex systems. Additionally, these methods may diverge or fail to find the appropriate solution under high loading levels. The Newton-Raphson (NR) method converges quadratically for initial estimates within the attraction region [32]. This can be crucial in situations where the region is narrow. Ideally, we need a robust method to search for the solution even when starting points are far from the solution.

During a real emergency, the network's condition can change so much that a traditional estimate-based approach may not be enough to locate the new operating point using a power flow solver. Therefore, it is important to develop solvers for large-scale systems that can work independently of the initial voltage estimate, for example, by starting from a flat voltage start guess.

2.3. The Homotopy Method

Considering a network scenario, previously known initial estimates may not be sufficient for the convergence of a PFP. The homotopy method can build a controlled path towards convergence in this context, working as a tool to solve nonlinear equations and find the roots of $\mathbf{g}(\mathbf{x}) = 0$ [5,22,36]. In the homotopy path, the intermediate results are used as initial estimates for the computation of states in the following points. So, it is possible to build a path coordinated by parameters until the final equilibrium state, characterizing the original problem solution. In fact, through a parameter h , the problem of a “difficult” solution at $h = 1$ is modeled for a system of an “easy” solution at $h = 0$. Once the solution is found by numerical resources in the easy solution, the parameter h evolves in the interval $h \in [0, 1]$, building the homotopy path until solving the difficult problem.

A static homotopy method was developed in [12] where two homotopy parameters were used. The h_1 parameter controls the insertion of fictitious elements in the network that cause changes in the main diagonal of the Y_{bus} matrix. The process allows for obtaining an easy-to-solve system, which is modified to start with no power flowing between any two buses. The states in this initial point are represented by the flat start voltages given as an estimate of the original PFP. Similarly, a second parameter, h_2 , is used to reduce the impedances connecting some buses in the neighborhood of the slack bus. The method proved suitable for large-scale considering well- and ill-conditioned systems.

The purpose of this paper is also to use a homotopy approach. However, our idea is to modify the conventional homotopy problem using a dynamic technique in which the algebraic equations are converted to ODEs with the homotopy parameter being the independent variable t . Also, the initial condition for the ODEs is precisely the initial estimate $\mathbf{x}^{(0)}$ used in the traditional PFP.

3. The Dynamical Homotopy Approach

In this section, we derive the dynamical homotopy problem, provide a numerical strategy for solving it numerically, and describe procedures for initializing the problem, considering explicit and implicit integration methods.

Initially, we are interested in solving a problem of this type:

$$\mathbf{G}(\mathbf{x}, t) = t\mathbf{g}(\mathbf{x}) + (1 - t)\mathbf{g}_0(\mathbf{x}) = 0, \quad (12)$$

where $\mathbf{G}(\mathbf{x}, t)$ is embedded from the original PFP power mismatch equation $\mathbf{g}(\mathbf{x}) = 0$ (composed of the power balance Equations (1) and (2)), the homotopy parameter t and an initial equation system that characterize the network for $t = 0$, $\mathbf{g}_0(\mathbf{x}) = 0$.

In (12), the solution of the PFP is obtained when $t = 1$. However, starting from $t = 0$, note that for each discrete value t_k , a nonlinear system of the type $\mathbf{G}(\mathbf{x}, t_k) = 0$ needs to be solved until $t_k = 1$. It is expected that the results found for each t_k are the states near the ones of the next point in t_{k+1} until the last step $t_{k+1} = 1$. Using the NR method to calculate the next point in the homotopy path, it is reasonable to expect convergence of the classical NR method in this scenario, thus combining a static homotopy and classical methods [37]. In our specific aim, we intend to minimize the necessity of using the NR method or comparable alternatives for this particular circumstance. In this sense, we convert (12) to a set of differential equations in the independent parameter t for using an integration method. Initially this dynamic approach to solving ill-posed problems was applied in other areas of knowledge [38,39]. This dynamic homotopy approach to solving PFP problems was presented by [27]. A general dynamic form uses a scalar in a homotopy function to solve nonlinear equations in the time-dependent domain function to improve ill-conditioned systems with poor initial value [40,41].

Since (12) depends on $\mathbf{x}$ and t , it can be differentiated, giving

$$d\mathbf{G}(\mathbf{x}, t) = \frac{\partial \mathbf{G}}{\partial \mathbf{x}} d\mathbf{x} + \frac{\partial \mathbf{G}}{\partial t} dt = 0, \quad 0 \leq t \leq 1. \quad (13)$$

Then, dividing (13) by dt , the transformation of (12) into a dynamic form is obtained as [28,42]

$$\frac{d\mathbf{G}(\mathbf{x}, t)}{dt} = \frac{\partial \mathbf{G}}{\partial \mathbf{x}} \frac{d\mathbf{x}(t)}{dt} + \frac{\partial \mathbf{G}}{\partial t} = 0, \quad 0 \leq t \leq 1, \quad \mathbf{x}(0) = \mathbf{x}^{(0)}. \quad (14)$$

Therefore, the idea is to find the PFP solution by solving an initial value problem (IVP) defined by the ODE

$$\mathbf{G}_x \frac{d\mathbf{x}(t)}{dt} = -\mathbf{G}_t, \quad 0 \leq t \leq 1, \quad \mathbf{x}(0) = \mathbf{x}^{(0)}, \quad (15)$$

along the homotopy path curve (pathway) $\gamma(t)$, in the parameter t , $0 \leq t \leq 1$, with an initial condition (IC) given by $\mathbf{x}(0) = \mathbf{x}^{(0)}$, where $\mathbf{G}_x = \mathbf{G}_x(\mathbf{x}, t) = \frac{\partial \mathbf{G}(\mathbf{x}, t)}{\partial \mathbf{x}}$ and $\mathbf{G}_t = \mathbf{G}_t(\mathbf{x}, t) = \frac{\partial \mathbf{G}(\mathbf{x}, t)}{\partial t}$, and $\mathbf{x}^{(0)}$ is precisely the initial estimation established for the iterative PFP.

In the same way, the ODE (15) can be solved by traditional numerical integration methods. For this goal, three techniques are investigated, each one having its own particularity: two explicit solvers, the Forward Euler (FE) and the 2nd-order Runge-Kutta (RK2); and the implicit form Backward Euler (BE). To better detail the influence of each method in the solution of the ODE and consequently in the improvement of the convergence solution of $\mathbf{g}(\mathbf{x}) = 0$, the solvers are presented in the sequel.

The FE method iterate given in (9), and adjusted to the dynamic form (15), is as follows

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \Delta t [\mathbf{G}_x(\mathbf{x}_k, t_k)]^{-1} \mathbf{G}_t(\mathbf{x}_k, t_k), \quad k = 0, 1, \dots \quad (16)$$

The recurrence characterized in the expression (16) to compute the state $\mathbf{x}_{k+1}$ from the previous state $\mathbf{x}_k$ characterize the numerical integration method as of type explicit.

An implicit form represented by the BE is given by

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \Delta t [\mathbf{G}_x(\mathbf{x}_{k+1}, t_{k+1})]^{-1} \mathbf{G}_t(\mathbf{x}_{k+1}, t_{k+1}), \quad k = 0, 1, \dots \quad (17)$$

From (17), the scheme is implicit because the state $\mathbf{x}_{k+1}$ which should be computed appears in the left- and right-side of the expression, differently from the computation in the explicit case, such as (16). Therefore in (17), $\mathbf{x}_{k+1}$ is an unknown of a nonlinear system. As a matter of fact, a nonlinear system

$$[\mathbf{G}_x(\mathbf{x}_{k+1}, t_{k+1})] \mathbf{x}_{k+1} - [\mathbf{G}_x(\mathbf{x}_{k+1}, t_{k+1})] \mathbf{x}_k + \Delta t \mathbf{G}_t(\mathbf{x}_{k+1}, t_{k+1}) = 0, \quad k = 0, 1, \dots \quad (18)$$

in the unknown $\mathbf{x}_{k+1}$ need to be solved, given that $\mathbf{x}_k$ is known.

However, solving the nonlinear system (18) is impractical, since the objective is just to solve the other power flow problem that is also nonlinear. Then, an approximation of the result in (18) is proposed computing just one valor in (17) by the well-known fixed point iteration (FPI) method using

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \Delta t [\mathbf{G}_x(\hat{\mathbf{x}}_{k+1}, t_{k+1})]^{-1} \mathbf{G}_t(\hat{\mathbf{x}}_{k+1}, t_{k+1}), \quad k = 0, 1, \dots \quad (19)$$

where $\hat{\mathbf{x}}_{k+1}$ is an estimation of $\mathbf{x}_{k+1}$ for the instant t_{k+1} .

In our simulations, it is used $\hat{\mathbf{x}}_{k+1} = \mathbf{x}_k$ and just one iteration is employed. Therefore, the process starts with $\hat{\mathbf{x}}_1 = \mathbf{x}(0) = \mathbf{x}^{(0)}$ and $t_1 = \Delta t_0$. The solution could be refined by updating the computed estimation $\hat{\mathbf{x}}_1$ in the FPI operation. In this case, the LU factorization of the matrix $[\mathbf{G}_x(\hat{\mathbf{x}}_{k+1}, t_{k+1})]$ could be kept and utilized for the following iterations. On the other hand, the simulations shown that only one iteration is sufficient to obtain a successful result for the dynamic homotopy process, avoiding unnecessary computations.

The explicit form defined by RK2 iterate is defined as

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \frac{\Delta t}{2} [\mathbf{K}_1 + \mathbf{K}_2], \quad k = 0, 1, \dots \quad (20)$$

where

$$\begin{aligned} \mathbf{K}_1 &= -[\mathbf{G}_x(\mathbf{x}_k, t_k)]^{-1} \mathbf{G}_t(\mathbf{x}_k, t_k) \quad \text{and} \\ \mathbf{K}_2 &= -[\mathbf{G}_x(\mathbf{x}_k + \Delta t \mathbf{K}_1, t_{k+1})]^{-1} \mathbf{G}_t(\mathbf{x}_k + \Delta t \mathbf{K}_1, t_{k+1}). \end{aligned} \quad (21)$$

3.1. Definition of the Homotopy Problem Easy Solution

The definition of the function $\mathbf{g}_0(\mathbf{x})$ is one of the targets in the resolution of the homotopy problem. It will affect the pathway of the solutions to reach the desired result, which is to find the roots of $\mathbf{g}(\mathbf{x})$.

In (12), it was defined $\mathbf{G}(\mathbf{x}, t)$. The equation represents a simple way of handling the homotopy path from the easy solution $t = 0$ to the solution of the original problem at $t = 1$. Note that $\mathbf{g}_0(\mathbf{x})$ is the function to establish the easy solution. The fixed point vector homotopy function (FPV) [41] was selected to represent $\mathbf{g}_0(\mathbf{x})$. However, in this paper, it was modified to $\mathbf{g}_0(\mathbf{x}) = K(\mathbf{x} - \mathbf{x}^{(0)})$ by introducing a factor K and different from unity. This change will play a key role in addressing ill-conditioned power flow problems, as will be shown later.

Considering the flat voltage start, voltage values for all PQ-bus have magnitude 1 pu; all buses have phase zero; and the voltage magnitudes are kept constant in the controlled voltage buses (PV-type). Therefore, the initial voltage magnitude vector is $\mathbf{V}^{(0)} = [1 \ 1 \ \dots \ 1]^T$, assigned only to PQ buses since the voltage magnitude is unknown in this type of bus. As at the beginning of the procedure $t = 0$, the boundary condition for the Equation (12) gives $\mathbf{G}(\mathbf{x}, 0) = 0 \cdot \mathbf{g}(\mathbf{x}) + (1 - 0)\mathbf{g}_0(\mathbf{x})$, i.e.,

$$\mathbf{G}(\mathbf{x}, 0) = \mathbf{g}_0(\mathbf{x}) = K \begin{bmatrix} \theta^{(0)} \\ \mathbf{V} - \mathbf{V}^{(0)} \end{bmatrix} = \mathbf{0}. \quad (22)$$

From (22), given that in $t = 0$ the result is $\mathbf{V}(0) = \mathbf{V}^{(0)}$ and $\theta^{(0)} = 0$, it is easy to verify that there is a trivial solution in $t = 0$ for the states and the ODE's IC coincides with the initial estimate of the PFP. Remark that the factor K is redundant at this IC point in view of the trivial solution. Therefore, we believe that this trivial assumption in other publications, such as in [41], contributed to the assigning of a unitary value for K to values $t > 1$ as well. However, the assumption can be exploited in another way to improve the performance of the dynamic homotopy approach.

Now, differentiating the Equation (12) with relation to the state vector $\mathbf{x}$, we have $\frac{\partial \mathbf{G}}{\partial \mathbf{x}} = t \frac{\partial \mathbf{g}}{\partial \mathbf{x}} + (1 - t) \frac{\partial \mathbf{g}_0}{\partial \mathbf{x}}$. Knowing that $\frac{\partial \mathbf{g}}{\partial \mathbf{x}} \Big|_{\mathbf{x}(t)} = \mathbf{J}(\mathbf{x}(t))$ is the PFP Jacobian matrix and $\frac{\partial \mathbf{g}_0}{\partial \mathbf{x}} = \mathbf{KI}$,

$$\mathbf{G}_x(\mathbf{x}, t) = t\mathbf{J}(\mathbf{x}(t)) + (1 - t)\mathbf{KI}, \quad t = k\Delta t, \quad k = 0, 1, 2, \dots, \quad (23)$$

where $\mathbf{I}$ is the identity matrix; Δt is a time-step defined for the integration method, and

$$\mathbf{G}_t(\mathbf{x}, t) = \mathbf{g}(\mathbf{x}(t)) - \mathbf{g}_0(\mathbf{x}(t)). \quad (24)$$

Note that the term $(1 - t)\mathbf{KI}$, in Equation (23), acts in the sense to attenuate (or remove) the ill-conditioning of $\mathbf{J}(\mathbf{x}(t))$ for $t \neq 1$. The impact depends on t and the factor K .

3.2. Performance of the Solvers in the First Time-Step

Assume that the homotopy pathway starts at $t = 0$ with the initial state $\mathbf{x}(0) = \mathbf{x}^{(0)}$, defined by the ODE's IC. It then evolves through time instants t_k , spaced by a time-step Δt , where $k = 0, 1, 2, \dots, N$ and N is the number of time-steps, until $t_k = 1$. The procedure allows for variable time steps Δt_i along the pathway, which can result in reduced accuracy for large time steps. However, at each time step, at least one LU factorization needs to be performed, because it is necessary to perform a computation of the type $\Phi(\mathbf{x}^{(k)}, t_k) = -[\mathbf{G}_x(\mathbf{x}^{(k)}, t_k)]^{-1} \mathbf{G}_t(\mathbf{x}^{(k)}, t_k)$, which increases the computational cost.

Therefore, various aspects need to be analyzed in order to determine an appropriate method for computing the solution of the nonlinear equation systems $\mathbf{g}(\mathbf{x}) = 0$ at the end of the homotopy pathway.

Given a time-step Δt_0 , the homotopy pathway can diverge from the exact path point for the first time-step, at $t_1 = \Delta t_0$, depending on the choice of the $\mathbf{g}_0(\mathbf{x})$ and Δt_0 . The analysis below concerns the choice based on the fixed point function, FPV, as defined previously, i.e., $\mathbf{g}_0(\mathbf{x}) = K(\mathbf{x} - \mathbf{x}^{(0)})$. The first point, $t_1 = \Delta t_0$, on the pathway will be calculated to demonstrate that even though Δt_0 is very small, using an explicit integration solver at this time may not be appropriate, especially for ill-conditioned systems.

Let the FE solver be used to compute the states in the step $t_1 = \Delta t_0 \ll 1$. Then, for $\mathbf{x}(0) = \mathbf{x}^{(0)}$ and $t = 0$, $\mathbf{G}_x(\mathbf{x}(0), 0) = \mathbf{K}\mathbf{I}$ and $\mathbf{G}_t(\mathbf{x}(0), 0) = \mathbf{g}(\mathbf{x}^{(0)})$. It implies from (16) that the increment $\mathbf{x}(t_1) - \mathbf{x}^{(0)} = -\Delta t_0[\mathbf{K}\mathbf{I}]^{-1}\mathbf{g}(\mathbf{x}^{(0)}) = -\frac{\Delta t_0}{K}\mathbf{g}(\mathbf{x}^{(0)})$. i.e., the deviation $\Delta\mathbf{x}(0) = \mathbf{x}(t_1) - \mathbf{x}^{(0)}$ is proportional to the mismatch $\mathbf{g}(\mathbf{x}^{(0)})$. However, in the case of ill-conditioned systems, $\|\mathbf{g}(\mathbf{x}^{(0)})\| \gg 1$ and the ratio $\Delta t_0/K$ needs to be set in a very small value. On the other hand, this strategy leads to small deviations $\Delta\mathbf{x}(0)$, causing slow convergence, with stagnation or leading to early divergence of the homotopy problem. The error increases when using an explicit integration method that requires the calculation of intermediary points, such as RK2. This is because computations involving the mismatch $\mathbf{g}(\mathbf{x}^{(0)})$ are involved even more. Additionally, the error increases because it depends on partial results of the type $\mathbf{x}(0) + K_1\mathbf{g}(\mathbf{x}^{(0)})$, as in the case of RK2. Therefore, it is necessary to establish a proper method to calculate t_1 with a very small time-step.

Now, let the BE scheme be used to compute the states in the instant $t_1 = \Delta t_0 \ll 1$. From (17), the increment is $\mathbf{x}(t_1) - \mathbf{x}_0 = -\Delta t_0[\Delta t_0\mathbf{J}(\mathbf{x}(t_1) + (1 - \Delta t_0)\mathbf{K}\mathbf{I})^{-1}\mathbf{g}(\mathbf{x}(t_1))]$ and the unknown is $\mathbf{x}(t_1)$, which arises in the two side of the equation. Considering the time-step $\Delta t_0 \ll 1$, the fixed-point iteration technique is applied to the expression to obtain $\mathbf{x}(t_1)$. Assume that, in its i -th iterate, the unknown is the variable $\mathbf{x}^{(i)}(t_1)$. In this paper, to minimize computations involving the inverse of $\mathbf{G}_x$, we suggest calculating just one iteration i and assigning $\mathbf{x}^{(0)}(t_1) = \mathbf{x}^{(0)}$ to compute $\mathbf{x}^{(1)}(t_1)$, the final result of interest, for a simplified result. Then, the updated result yields

$$\begin{aligned}\Delta\mathbf{x}(0) &= -[\mathbf{J}(\mathbf{x}^{(0)}) + \frac{(1 - \Delta t_0)}{\Delta t_0}\mathbf{K}\mathbf{I}]^{-1}\mathbf{g}(\mathbf{x}^{(0)}) \approx -[\mathbf{J}(\mathbf{x}^{(0)}) + \frac{K}{\Delta t_0}\mathbf{I}]^{-1}\mathbf{g}(\mathbf{x}^{(0)}) = \\ &= -[\mathbf{J}(\mathbf{x}^{(0)}) + \delta\mathbf{I}]^{-1}\mathbf{g}(\mathbf{x}^{(0)}),\end{aligned}\quad (25)$$

where $\delta = K/\Delta t_0$.

Interesting to note that in (25), the original PFP ill-conditioned Jacobian $\mathbf{J}(\mathbf{x}^{(0)})$ of the initial iteration was modified by the introduction of the small-perturbation factor δ for this step t_1 . According to [29], the small perturbation factor allows an adequate conditioning procedure to solve the original ill-posed problem. Also, in the function of the assignment of δ and Δt_0 , the factor K is determined as $K = \delta\Delta t_0$. Therefore, not necessarily equal to unity.

Another form to achieve a good result when the step size t_1 is very small, given $\mathbf{x}^{(0)}$, is by solving (12) for a specific t_1 . In this scenario, only the states are considered as variables. Then, an approximation for the states at t_1 can be obtained by solving the algebraic equation $\mathbf{G}(\mathbf{x}) = \Delta t_0\mathbf{g}(\mathbf{x}) + (1 - \Delta t_0)K(\mathbf{x} - \mathbf{x}^{(0)}) = 0$, assuming a linear approximation for $t_1 \ll 1$. According to [28], a linear approximation of $\mathbf{G}(\mathbf{x})$ around $\mathbf{x}^{(0)}$ gives

$$\mathbf{G}(\mathbf{x} + \Delta\mathbf{x}) \approx \mathbf{G}(\mathbf{x}^{(0)}) + \mathbf{G}_x(\mathbf{x}^{(0)})\Delta\mathbf{x} = 0. \quad (26)$$

Considering that $t_1 = \Delta t_0$ for computing (26), then $\mathbf{G}(\mathbf{x}^{(0)}) \neq 0$, $\mathbf{G}_x(\mathbf{x}^{(0)}) = \Delta t_0\mathbf{J}(\mathbf{x}^{(0)}) + (1 - \Delta t_0)K\mathbf{I}$, and $\mathbf{G}(\mathbf{x}^{(0)}) = \Delta t_0\mathbf{g}(\mathbf{x}^{(0)}) + (1 - \Delta t_0)K(\mathbf{x}^{(0)} - \mathbf{x}^{(0)}) = \Delta t_0\mathbf{g}(\mathbf{x}^{(0)})$. Then, the linear approximation in (26) gives the deviation

$$\Delta\mathbf{x} = \mathbf{x} - \mathbf{x}^{(0)} = -[\mathbf{G}_x(\mathbf{x}^{(0)})]^{-1}\mathbf{G}(\mathbf{x}^{(0)}) = [\Delta t_0\mathbf{J}(\mathbf{x}^{(0)}) + (1 - \Delta t_0)K\mathbf{I}]^{-1}\Delta t_0\mathbf{g}(\mathbf{x}^{(0)}). \quad (27)$$

Note that, after some algebraic handling, the result in (27) is identical to that in (25), and the result obtained by the linear approximation and through the BE solver agrees. Hence, using these two forms for approximating the states at the time step $t_1 = \Delta t_0$ provides results close to the homotopy pathway. These approximations are suitable to be used as initial conditions at t_1 for numerical integration methods for the homotopy pathway. Particularly, explicit techniques are unable to yield satisfactory results without this initialization.

The remaining points, for $t_1 < t_k \leq 1$, should preferably be calculated with integration steps Δt much larger than the initial one Δt_0 . This requisite is reinforced by the need to minimize the amount of computations of $\Phi(\mathbf{x}(t_k), t_k)$ at each step t_k . However, explicit methods may be more sensitive to the high integration step. On the other hand, an implicit method requires iterations to obtain the result in the integration step itself and admits a large time-step. In one way or another, the goal is to minimize performing LU factorizations until the final result at $t = 1$ is reached, as these factorizations require the highest computational costs. In explicit methods of the FE and RK2 types, at least 1 LU factorization is performed to obtain the result in one step. In the implicit BE method, only 1 LU iteration is proposed to be used in this work to obtain the result. However, this approach yields an approximate result for the step due to the need for more iterations to achieve high precision. It is suggested that the results of the homotopy problem at step $t = 1$ in this work may be inaccurate with respect to the error requirements needed for the PFP, based on the characteristics of the methods. The result obtained from the integration process is only considered to be a rough estimate of the exact solution to the load flow problem. Hence, a conventional method such as NR or its fast decoupled form, FDXB (see [1,30] for details of the decoupled method), is a good approach to improve the solution obtained from dynamic homotopy.

Therefore, a compromise must be sought between the expected result at the end of the dynamic homotopy simulation horizon and the expected accuracy of the PFP solution. Considering that it is not reasonable to use an exaggerated number of time-steps in the simulation because the dynamic homotopy process would be very computationally expensive, an alternative would provide an approximation of the result in time $t = 1$. However, at this point, an approximate solution for the states may be unaccepted and would need to be refined. In simpler terms, in this scenario, the homotopy method changes the initial estimate $\mathbf{x}^{(0)}$ to a new state $\hat{\mathbf{x}}^{(0)} = \mathbf{x}(t)|_{t=1}$. The transformed state is expected to be within an attraction region and should give a good starting point for convergence, even for a numerical method like the classical NR and FDXB methods.

The idea is to use the dynamic homotopy method to determine an approximate solution for the PFP, starting from a flat start estimate, equivalent to an improvement in the region of attraction in the original problem. From this result, the PFP solution can be calculated with high precision. Furthermore, this new starting point, $\mathbf{x}(t)|_{t=1}$, can be used by other iterative techniques different from the traditional NR.

3.3. A Generic Tutorial Example

To better understand the application of the concepts, we first examine a tutorial example from [29] before applying the details to large-scale ill-conditioned power system models. The problem consists of finding the roots of the nonlinear equation generic system represented by $g_1(x_1, x_2) = x_1^2 + x_2^2 - 2x_1x_2 - 1 = 0$, $g_2(x_1, x_2) = x_1 + x_2 - 2 = 0$, $\mathbf{g}(x_1, x_2) = [g_1(x_1, x_2) \ g_2(x_1, x_2)]^T$, and $\mathbf{x} = [x_1 \ x_2]^T$.

In order to iteratively solve the system, an initial estimate $\mathbf{x}^{(0)} = [1 \ 1]^T$ is assigned. Note that at this point, the Jacobian matrix is singular. However, in the neighborhood of $\mathbf{x}^{(0)}$, the singularity disappears for other points. Then, to eliminate the singularity at $\mathbf{x}^{(0)}$, it was proposed in [29] to replace one of the variable values with the same value incremented by a small value ϵ , for example, $x_2^{(0)} = 1 - \epsilon$. The Newton-Raphson solver has been found to not converge without the added perturbation. Additionally, achieving convergence may be difficult depending on the specific value of ϵ . However, the dynamic homotopy-

based technique addresses this issue by incorporating the explicit and implicit integration methods we have previously discussed, along with any necessary adjustments.

Two different approaches were used for simulations: one using only the classical Newton-Raphson (NR) scheme, starting from $\mathbf{x}^{(0)}$; and the another using a combination of the dynamic homotopy-based method and the NR solver. By this latter, the NR solver uses $\hat{\mathbf{x}}^{(0)} = \mathbf{x}(t)|_{t=1}$ as the initial guess, which was obtained previously by a dynamic homotopy numerical scheme.

When the NR is used starting directly from $\mathbf{x}^{(0)}$, it was investigated with different values of ϵ to modify only the entry $\mathbf{x}_2^{(0)}$.

In the hybrid approach, an integration method is initially used to calculate a partial result at $t = 1$, generating $\hat{\mathbf{x}}^{(0)} = \mathbf{x}(t)|_{t=1}$. This partial result is used as a guess for the classical NR solver to compute the high-accuracy solution $\mathbf{x}_*$.

The integration methods FE, BE, and RK2, were used as proposed previously. In all dynamic homotopy simulations, the initial time step $\Delta t_0 = 0.01$ produces the first step $t_1 = 0.01$. The subsequent steps $t_k > t_1, k = 2, 3, \dots$ were determined by a constant time step, denoted as Δt . Two different types of simulations were run with constant values for Δt : 0.5 and 0.125. In addition, the value of K was varied, with the following values: 0.005, 0.01, and 0.05. For all simulations using dynamic homotopy, the results were initially computed at time t_1 using only the BE solver, or equivalently, by using the linear approximation for the static homotopy equation. Then, the states obtained at time t_1 were used as the starting point for the FE, BE, and RK2 numerical integration solvers for $t_k > t_1, k = 2, 3, \dots$ with a given Δt . The Figure 1 illustrates the mismatch plots for the simulations. In Figure 1a, the plots for results were obtained using only the NR solver, while the other subfigures exhibit the plot visualizations for the hybrid approach, considering constant step-time $\Delta t = 0.5$. Convergence was achieved for all simulations and, after using the NR solver, reached the refined values $x_{1*} = 1.5$ and $x_{2*} = 0.5$ with a tolerance for the mismatch of 10^{-5} .

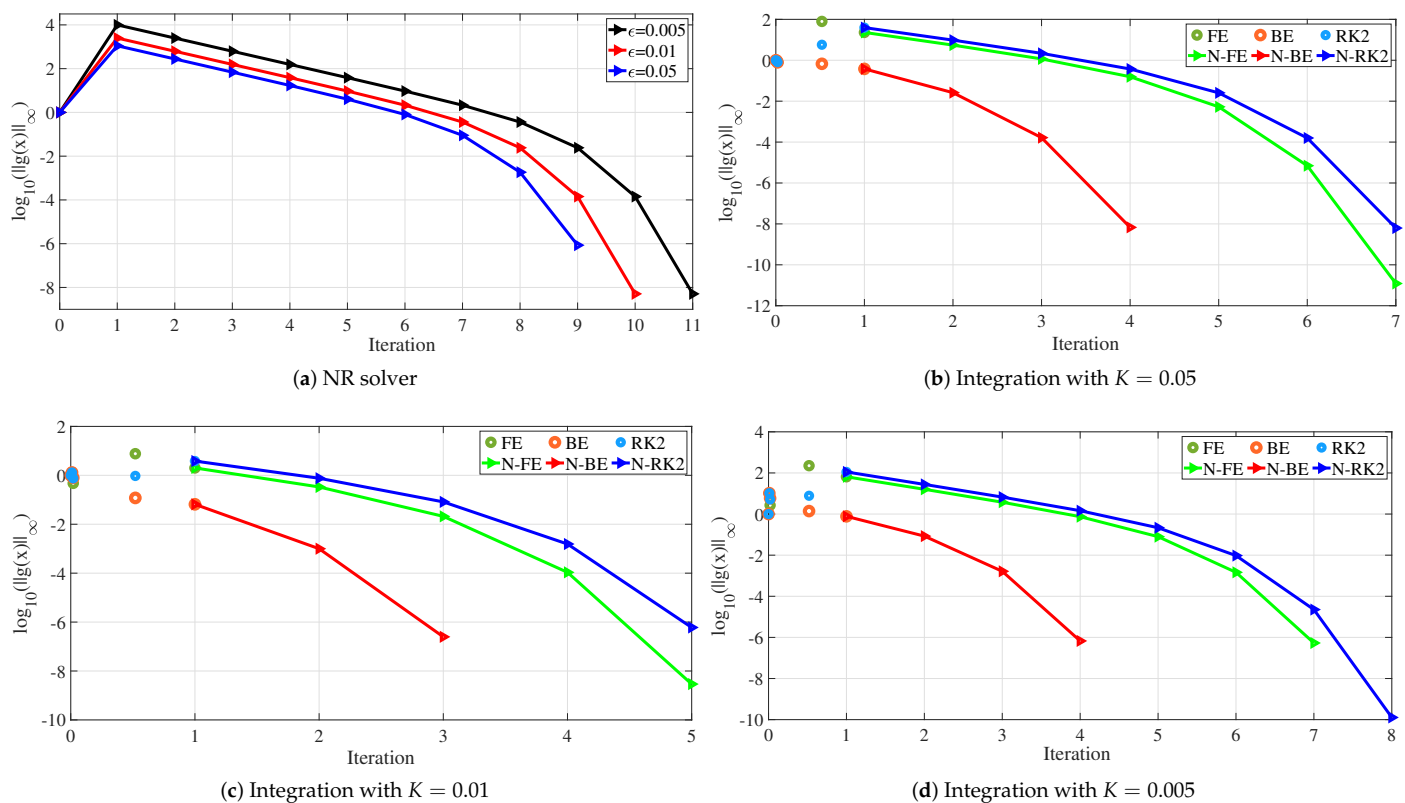


Figure 1. Results determined by using only NR and a combination of dynamic homotopy, with $\Delta t = 0.5$, and the NR solver.

The analysis of the results shows that when the NR solver departs directly from the original estimate $\mathbf{x}^{(0)}$ with $x_2^{(0)} = 1 + \epsilon$, the mismatches initially experience a sudden increase for any perturbation value ϵ , and then gradually decrease with each iteration. A minimum of 9 to 11 iterations were necessary to achieve a tolerance of 10^{-5} in these experiments. In the hybrid approach (refer to Figure 1b–d)), the NR solver utilizes the result obtained for the point when $t = 1$. The discrete intermediate points in the plots for the range 0 to 1 refer to results obtained from the dynamic homotopy solvers. With the enhancements introduced by the homotopy solvers, the Newton-Raphson solver exhibits improved convergence behavior, consistently achieving convergence within a maximum of 8 iterations across a range of values for the parameter K . This enhanced performance is observed when the NR solver is utilized in conjunction with the Forward Euler (FE), Backward Euler (BE), and RK2 solvers. In all cases, the best performance was observed with the BE (implicit form) solver, which required no more than 3 iterations to reach a solution. The explicit forms (FE and RK2) also achieved the solution with fewer iterations than the NR solver properly when starting directly from $\mathbf{x}^{(0)}$.

Another set of experiments was accomplished to verify the influence of the time-step Δt , keeping the same step initial, $t_1 = 0.01$, and using $K = 0.05$. For this case, a constant value $\Delta t = 0.125$ was set. The Figure 2 depicts plots for the hybrid approach involving the experiments.

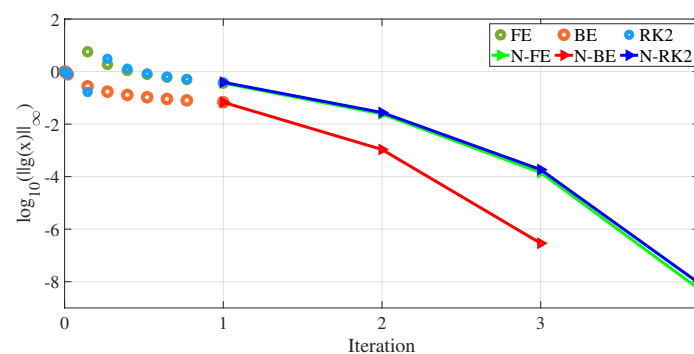


Figure 2. Integration solver performance when a smaller time-step $\Delta t = 0.125$ was adopted after the step $t_1 = 0.01$.

From Figure 2, we can confirm that the behavior observed in the results shown in Figure 1 was maintained. However, the explicit solvers now require fewer iterations to achieve a highly accurate solution using the Newton-Raphson method. The backward Euler solver did not show a significant improvement compared to its performance with a high time-step size, but it still outperformed the other dynamic homotopy solvers.

The upcoming section discusses experiments and results on various poorly conditioned large-scale power system models. The goal is to show how effective the proposed method is in accurately finding the solution of the PFP by utilizing the desired performance of the dynamic homotopy-based technique.

4. Tests and Results

This section describes experiments and results associated to the technique investigated in this work. The simulations were performed by using a modified version of MATPOWER v6.0 to incorporate the proposed methodology. A tolerance of 10^{-8} pu for power mismatch was established for checking PFP convergence accuracy. All initial PFP estimations are of the flat start type, except for one from the MATPOWER databank used as a reference for validating the results.

In the experiments, all simulations had two parts. First, the PFP solution was determined using the dynamic homotopy method to find a preliminary solution $\hat{\mathbf{x}}^{(0)}$. The states obtained in this preliminary solution were calculated using FE, BE, and RK2 solvers. In

the final part, the preliminary solution was refined employing the classical NR and FDXB - fast-decoupled power flow (XB version, according to [30]) algorithms.

For reference, all simulations were performed on a personal computer with Intel(R) Core(TM) i7-8565U 8th Gen, 256 GB SSD CPU, 1.80 GHz, 8 GB RAM, 64-bit OS. The MATLAB version was R2018a with MATPOWER 6.0 available at [43].

4.1. Test Systems

Experiments were performed to validate the proposed method in systems with two different characteristics: one with an ill-conditioned profile and another under stressed conditions. The models were obtained from [44]. They were built using a combination of existing cases from MATPOWER's databank [30]. They are derived from the files `case9241pegase.m` and `case13659pegase.m` that represent European high voltage transmission network with 9241 and 13,659 buses respectively, that operates at 750, 400, 380, 330, 220, 154, 150, 120, and 110 kV [45]. The Table 1 shows the list of the ill-conditioned cases used from [44], whose further details can also be found in [46].

Table 1. Ill-conditioned test systems according to new scenarios and originals [46] and the main physical characteristics.

Ill-Conditioned Case	Modified Case	Number of Elements					
		Buses	Generators	Loads	Shunts	Branches	Transformers
<code>case18482.mat</code>	<code>case9241pegase</code> $\times 2$	18,482	2890	9790	14,654	32,113	2638
<code>case27318.mat</code>	<code>case13659pegase</code> $\times 2$	27,318	8184	11,088	17,508	40,935	11,426
<code>case36964.mat</code>	<code>case9241pegase</code> $\times 4$	36,964	5780	19,580	29,308	64,227	5276
<code>case54636.mat</code>	<code>case13659pegase</code> $\times 4$	54,636	16,368	22,176	35,016	81,871	22,852
<code>case109272.mat</code>	<code>case13659pegase</code> $\times 8$	109,272	32,736	44,352	70,032	163,743	45,704

The cases involving scenarios where a load limit is identified (stressed conditions) are labeled as `case69limit.mat`, `case141limit.mat`, `case_ACTIVSg500limit.mat` and `case_ACTIVSg2000limit.mat`. They were obtained from [47]. In that work, they were constructed by modifying existing cases from the original loading scenarios in the MATPOWER database. The changes were performed in the active and reactive power injected into each load bus up to the standard NR convergence limit, always starting from a flat voltage start. The previous iteration of change in this power of 0.0001 pu, which would lead to divergence, was considered the load limit state.

4.2. Simulations to Assess the Performance of the Integration Solvers

In this subsection, several experiments are assessed to demonstrate the performance of the dynamic homotopy solver's contributions to improving the convergence of the classical NR solver. The studies initially prioritized the characteristics of integration methods, focusing on the influence of the time-step size.

All simulations involving the integration methods require a small time-step to compute the initial step in $t_1 = \Delta t_0$. This dependence can be physically explained by the smallest "time constant" associated with the system's dynamics under study. For the simulations in this work, the initial time-step, Δt_0 , is a factor for investigations since ensuring an adequate result for the dynamical homotopy-based technique is a key aspect for evaluating the method's efficiency. Therefore, the discrete points in the path curve, $\gamma(t)$, $0 \leq t \leq 1$, are $t_0 = 0$, $t_1 = \Delta t_0$, and $t_{k+1} = \Delta t_0 + k\Delta t_k$, $k = 1, \dots, (N-1)$. The initial point at $t = 0$ is the IC with states $\mathbf{x}(0) = \mathbf{x}^{(0)}$, which also coincides with the original starting point for the PFP. N represents the number of points beyond 0 in the homotopy pathway and should be restricted to prevent situations for $t_k > 1$. Therefore, if necessary, the final point t_N must be set in the unitary value and the time-step properly reduced. For instance, in the extreme case where $\Delta t_0 = 0.005$ and $\Delta t_1 = 1.0$ were specified, the definition of the time instants t_k (or steps) would be as follows: $t_1 = \Delta t_0 = 0.005$, $t_2 = \Delta t_0 + \Delta t_1 = 1.005$, with

$N = 2$ points plus $t_0 = 0$. However, $t_2 > 1$, and the point needs to be readjusted to give $t_2 = 1$. In this case, the time-step must be reduced to $\Delta t_1 = 0.995$ instead of $\Delta t_1 = 1.0$. The variable N represents the number of points on the homotopy pathway where the states $\mathbf{x}(t_k)$, $k = 1, \dots, N$ need to be computed. Using a numerical integration method that minimizes N is beneficial because computing each point requires an LU factorization of the matrix $\mathbf{G}(\mathbf{x}(t_k))$ (refer to Equations (25) and (27) for more details). This aspect becomes more evident in the following simulations for computing states only at the instant t_1 , which suggests the use of an implicit method for ill-posed systems.

4.2.1. Experiments for the First Time-Step t_1

In the context of the dynamic homotopy scheme, it is important to determine the states for the initial step when $t_1 = \Delta t_0$. This evaluation impacts the setting of the factor K in the definition of the “easy” function $\mathbf{g}_0(\mathbf{x})$ and, consequently, the conditioning of the state derivative matrix, $\mathbf{G}_x(\mathbf{x}, t)$.

Assume that the PFP is of the type ill-conditioned. In reason of the nature of this type of nonlinear system, the mismatches for the initial estimate generally present $\|\mathbf{g}(\mathbf{x}^{(0)})\| \gg 1$. At the beginning of the homotopy pathway, the starting point is initially known due to IC and given by $t_0 = 0$, $\mathbf{x}(0) = \mathbf{x}^{(0)}$. Then, the states $\mathbf{x}(t_1)$ in the point defined by $t_1 = \Delta t_0$ can be calculated in the sequence by $\mathbf{x}(t_1) = \mathbf{x}(t_1) + \Delta \mathbf{x}(0)$. Considering the dynamical homotopy-based problem, a numerical integration technique should be used to determine $\Delta \mathbf{x}(0)$.

Based on the first approximated states, $\mathbf{x}(t_1)$, calculated at t_1 , divergence along the homotopy pathway may already be verified. Then, it is fundamental to study the numerical behavior of the states for this point. The states, $\mathbf{x}(t_1)$ states were calculated for a small Δt_0 following this strategy. For these simulations, the initial time-step was fixed at 0.005, giving $t_1 = 0.005$. To demonstrate the inefficacy of explicit integration methods in this situation, we applied the FE and BE solvers to compute the states at t_1 . The value 0.0001 was assigned for K . Initially, as presented in Section 3.2 and considering the FE solver, the deviation concerning the initial states, $\mathbf{x}(0) = \mathbf{x}^{(0)}$, is $\Delta \mathbf{x}(0) = \mathbf{x}(t_1) - \mathbf{x}(0) = -(0.005/0.0001)\mathbf{g}(\mathbf{x}^{(0)}) = -50\mathbf{g}(\mathbf{x}^{(0)})$. This result demonstrates that the numerical pattern of $\|\Delta \mathbf{x}(0)\|$ contributes to amplifying and propagating the characteristics of the ill-conditioning of the original PFP. This is justified because, for ill-conditioned systems, as stated before, $\|\mathbf{g}(\mathbf{x}^{(0)})\|$ is generally very large. Then, it is unacceptable to compute the state for the instant t_1 as $\mathbf{x}(t_1) = \mathbf{x}(0) - 50\mathbf{g}(\mathbf{x}^{(0)})$, as it will cause a numerical explosion of the norm $\|\mathbf{g}(\mathbf{x}(t_1))\|$. On the other hand, incrementing K , for example, setting it into a unitary value, results in small increments $\Delta \mathbf{x}(0) = \mathbf{x}(t_1) - \mathbf{x}(0) = -0.005\mathbf{g}(\mathbf{x}^{(0)})$. However, this result yields a greatly reduced variation $\Delta \mathbf{x}(0)$, proportional to mismatches, likely without significantly changing the states in t_1 . When using the RK2 scheme, the result becomes even worse because not only do we need to directly evaluate the states at t_1 to compute K_1 (as required for FE), but it is also needed an additional evaluation to calculate K_2 using an inadequate computed K_1 (refer to Equation (21)). Therefore, we conclude that using FE or RK2 (explicit integration techniques) would significantly increase the error, rendering the solution ineffective for computing the states across the homotopy pathway.

In contrast, when considering the BE solver (or even a linear approximation of the homotopy function as calculated by using (27)) at t_1 , the deviation $\Delta \mathbf{x}(0)$ is calculated using (25). Using the same values adopted for the explicit integration analysis, $K = 0.0001$ and $\Delta t_0 = 0.005$, a small scaling factor increment $\delta = K/\Delta t_0 = 0.0001/0.005 = 0.02$ is calculated, and provides a “perturbation term” $\delta \mathbf{I}$ to be added to the diagonal of the initial PFP Jacobian matrix in (25). Note that the Jacobian matrix is the same as the one for the first iteration of a standard iterative technique, such as the NR scheme. Therefore, the adjustment of δ is a function of K and Δt_0 . So, this procedure is expected to effectively mitigate the ill-conditioning of the Jacobian matrix during the dynamic simulation at time t_1 . However, only the setting of δ is insufficient since large Δt_0 can also make the convergence process infeasible, as shown in simulations later. Therefore, the investigations considering

these aspects should be considered to achieve results for successfully resolving the dynamic homotopy problem for time instants $t_k > t_1$, even when Δt_k is substantial. Ultimately, this will establish a robust starting point for the NR solver and other iterative schemes.

4.3. Time-Step Change Study

Based on the previous findings and descriptions, experiments were carried out using the FE, RK2, and BE solvers, starting from the point t_1 instead of $t_0 = 0$. This procedure aligns with adopting only an implicit integration technique to compute the states initially in the instant t_1 , according to the description forwarded in Section 4.2.1. The value $K = 0.0001$ was adopted in the simulations.

Therefore, it was assumed that the simulations must start from the point whose states $\mathbf{x}(t_1)$ were calculated at time t_1 using the initial point $t_0 = 0$, where the IC is $\mathbf{x}(0) = \mathbf{x}^{(0)}$. For this previous result, only the BE solver or a linear approximation of the homotopy problem (see (27)) was used to obtain it.

Then, from this part of the paper, the adopted initial conditions (ICs) for FE, RK2, and BE solvers are considered starting in the point defined in the homotopy pathway by t_1 , $(\mathbf{x}(t_1), t_1)$.

The numerical integration solvers calculate points with $t_k > t_1$, $k = 1, \dots, N - 1$. To evaluate the performance of the three numerical integration solvers, initially simulations with $t_2 = 2\Delta t_0$ was assumed. Experience has shown that setting t_2 in large time steps, unless using the implicit method, results in premature convergence failures of the explicit methods.

To compare the performance of the implicit method (BE) with the explicit methods (FE and RK2), we carried out simulations using two different types of time steps for $t_k > t_2$. In the first set of experiments, we used an elevated constant time-step, unless to compute the last point in $t_N = 1$. In the second set, we modified it to be a variable time-step that increases over time.

The Tables 2–4 exhibit simulation results, respectively, for the solvers FE, RK2, and BE, when $\Delta t_0 = 0.005$ (used for determining t_1 and t_2) is adopted; and $\Delta t = 0.5$ (large time-step), used for $t > t_2$. The second row in the tables indicates either the discrete-time for the dynamic homotopy calculations (2nd to 6th columns) or the iterations for the NR solver (columns 7th to 9th), which uses the result in $t = 1$ of the dynamic homotopy problem, i.e., $\hat{\mathbf{x}}^{(0)}$, as an initial estimate. Depending on the solver used, the results of the infinite-norm of the PFP mismatches, $\|\mathbf{g}(\mathbf{x})\|_\infty$, are shown in 2nd to 9th columns, based on the case defined in the 1st column.

Table 2. Norm evolution for the FE solver combined with the NR solver for an initial time-step $\Delta t_0 = 0.005$ and large time-step for $t > 0.01$.

CASE/Time		$\ \mathbf{g}(\mathbf{x})\ _\infty$ of the Solver					$\ \mathbf{g}(\mathbf{x})\ _\infty$ NR	
t or $iter$	0.0	0.005	0.01	0.51	1.0	1	2	3
case18482	532	65	9.7	2×10^3	2×10^3	-	-	-
case27318	200	25	1.2	27	2×10^7	-	-	-
case36964	532	65	3.5	2×10^3	3×10^5	-	-	-
case54636	200	25	1.2	27	1×10^9	-	-	-
case109272	200	25	1.2	27	1×10^9	-	-	-
case69limit	0.012	2×10^{-4}	1×10^{-5}	5×10^{-4}	2×10^{-5}	2×10^{-8}	3×10^{-12}	-
case141limit	0.006	2×10^{-4}	9×10^{-6}	5×10^{-4}	2×10^{-5}	4×10^{-8}	1×10^{-11}	-
case500limit	13	1.2	0.02	0.9	1.0	8×10^{-3}	2×10^{-6}	4×10^{-13}
case2000limit	54	3.9	0.8	210	7×10^3	-	-	-

Table 2 refers to a significant time-step immediately following the instant $t_2 = 0.01$. From the computed norms, it is verified that the FE solver is unable to furnish an adequate estimate $\hat{\mathbf{x}}^{(0)}$ for the NR solver in $t = 1$, when employed to solve the dynamic homotopy

problem for the large-scale ill-conditioned base cases. However, when applied to small systems, even considering the loading in the limit (69-, 141-, and 500-bus cases), it provided an appropriate starting, which converged with no more than three additional iterations. However, a failure of convergence occurred for the 2000-bus system. This is the largest model among those with studied stressed conditions.

Table 3. Norm evolution for the RK2 solver combined with the NR solver for an initial time-step $\Delta t_0 = 0.005$ and large time-step for $t > 0.01$.

CASE/Time		$\ g(x)\ _\infty$ of the Solver				$\ g(x)\ _\infty$ NR		
t or $iter$	0.0	0.005	0.01	0.51	1.0	1	2	3
case18482	532	65	33	3×10^7	1×10^{11}	-	-	-
case27318	200	25	13	6×10^5	1×10^{10}	-	-	-
case36964	532	65	33	3×10^9	2×10^{14}	-	-	-
case54636	200	25	13	7×10^5	1×10^{10}	-	-	-
case109272	200	25	13	1×10^7	5×10^{12}	-	-	-
case69limit	0.012	2×10^{-4}	1×10^{-4}	9×10^{-5}	5×10^{-5}	1×10^{-9}	-	-
case141limit	0.006	2×10^{-4}	8×10^{-5}	4×10^{-4}	2×10^{-4}	7×10^{-9}	-	-
case500limit	13	1.2	0.6	1×10^4	5×10^3	-	-	-
case2000limit	54	3.9	1.9	1×10^4	4×10^4	-	-	-

The explicit technique implemented for the RK2 solver performs similarly to the FE solver. However, in some cases (see Table 3), the performance is even worse, as there was no convergence for smaller systems. This behavior is because RK2 requires an intermediate LU factorization to reach the instant of interest, according to the defined time-step Δt . Since the method reaches large errors in $t = t_2 = 0.01$, these errors propagate and lead to erroneous results in the following steps. From Tables 2 and 3, in $t_3 = 0.51$, results with a norm of at least 10^4 are observed. Although the FE solver also shows erratic computation, the norms are smaller than those found by RK2 solver (for reference, see the 5th column in both tables). However, neither FE nor RK2 performed satisfactorily and gave adequate starting for the NR solver.

The results shown in Table 4 indicate that the BE method successfully converged for all test cases. This demonstrates that the single implicit method is suitable for solving the dynamic homotopy problem, even when using elevated time steps. When initiated with the results obtained by the dynamic homotopy-based technique, the NR solver required at most three iterations for convergence, regardless of the system studied.

Table 4. Norm evolution for the BE solver combined with the NR solver for an initial time-step $\Delta t_0 = 0.005$ and large time-step for $t > 0.01$.

CASE/Time		g(x) _∞ of the Solver					g(x) _∞ NR	
<i>t</i> or <i>iter</i>	0.0	0.005	0.01	0.51	1.0	1	2	3
case18482	532	65	33	7.2	3.7	0.04	1.5×10^{-5}	1×10^{-11}
case27318	200	25	13	0.5	0.25	0.1	2.5×10^{-5}	7×10^{-12}
case36964	532	65	33	6.9	3.6	0.27	6×10^{-4}	4×10^{-9}
case54636	200	25	13	0.5	0.25	0.1	3×10^{-5}	7×10^{-12}
case109272	200	25	13	0.5	0.25	0.1	3×10^{-5}	7×10^{-12}
case69limit	0.012	2×10^{-4}	1×10^{-4}	2×10^{-6}	5×10^{-7}	5×10^{-12}	-	-
case141limit	0.006	2×10^{-4}	8×10^{-5}	1×10^{-6}	5×10^{-7}	6×10^{-11}	-	-
case500limit	13	1.2	0.6	0.02	9×10^{-3}	1×10^{-6}	2×10^{-13}	-
case2000limit	54	3.9	2	2.8	1.5	0.01	8×10^{-7}	9×10^{-13}

In the upcoming experiments, the methods were assessed, increasing time steps for $t_k > t_2$. The specific time points $t_1 = 0.005$, $t_2 = 0.01$, $t_3 = 0.02$, $t_4 = 0.12$, $t_5 = 0.62$, and $t_6 = 1.0$ were established. It should be noted that the time step $\Delta t_4 = t_5 - t_4 = 0.5$

is considered large. Again, the infinite norms of the mismatches were observed for the dynamic homotopy and NR solvers. Considering the similar profiles verified for the mismatch norms of case109272, case54636, and case27318 and based on the findings in Tables 2–4, only the results for the largest system is shown for these specific experiments, i.e., the case109272. Similarly, the same remark is made for the experiments involving the case36964 and case18482, being considered only the former. Just the largest system was considered concerning the limit loading cases. Given the explanations regarding the selected systems, the new simulation results are presented in Table 5.

In Table 5, the mismatch norm was computed for each point defined by a specific time instant t_k , according to the employed solver. Also, only the result for three iterations of the NR solver for refining the result is exhibited in the table (see 10th to 12th columns).

Table 5. Evolution of norms with reduced initial time steps and progressive increments in computing the dynamic homotopy results, followed by three iterations of the NR solver.

CASE	Solver/Time	$\ g(x)\ _\infty$ of the Solver							$\ g(x)\ _\infty$ for NR		
	t or $iter$	0.0	0.005	0.01	0.02	0.12	0.62	1.0	1	2	3
case36964	FE	532	65	3.5	0.7	5.7	57	114	28	5	0.2
	RK2	532	65	33	16	9.6	4.2	2.6	0.2	3×10^{-4}	1×10^{-9}
	BE	532	65	33	16	4.0	1.4	0.9	0.09	9×10^{-5}	1×10^{-10}
case109272	FE	200	25	1.2	0.02	0.07	0.3	0.17	0.03	6×10^{-4}	2×10^{-9}
	RK2	200	25	13	6.3	1.2	0.5	0.3	0.19	2×10^{-4}	2×10^{-9}
	BE	200	25	13	6.3	1.1	0.2	0.1	0.02	1×10^{-6}	1×10^{-11}
case2000limit	FE	55	3.9	0.8	0.7	3.6	58	91	16	1.0	6×10^{-3}
	RK2	55	3.9	1.9	1.0	5.0	5.5	3.4	0.2	3×10^{-4}	1×10^{-9}
	BE	55	3.9	1.9	1.0	1.0	0.3	0.2	5×10^{-4}	3×10^{-9}	-

The results from Table 5 show that all solvers now exhibit favorable convergence performance, except for the explicit FE solver in the case36964 and case2000limit. In fact, they achieved convergence after two and one more additional iteration, respectively. The BE solver once again performed well in obtaining an initial result for the NR solver at $t = 1$. The RK2 solver also performed well, almost as well as BE, but with slightly lower performance in the system with high loading.

One drawback of the experiments highlighted in Table 5 is that additional computations were required because two more time points t_k were necessary compared to the first part of the experiments shown in Tables 2–4. Despite this, no significant improvement was observed in reducing the number of iterations in the refinement process using the NR solver. It is important to note that a new LU factorization of the Jacobian matrix is required at each time point added, increasing the computational cost of the process.

4.4. Changes in the Time-Step and Factor K

In the Section 4.3, the simulations were carried out using the parameters $K = 0.0001$ and $\Delta t_0 = 0.005$. In the present subsection, new experiments are conducted modifying these parameters to evaluate their impact on the performance of the dynamical homotopy-based technique. Both the initial time-step Δt_0 and the factor K , the latter composing the “easy” function $g_0(x)$, were multiplied by a scaling factor ten when compared to the previous values. So, the new values are $\Delta t_0 = 0.05$ and $K = 0.001$. This way, the perturbation parameter in (25) $\delta = K/\Delta t_0 = 0.02$ was kept in the same value despite different values for Δt_0 and K . Still, the dynamic of the time simulation due to the numerical integration process was modified. The upcoming investigations aim to confirm whether the approximate result $\hat{x}^{(0)}$ produced by the dynamical homotopy solver is appropriate for initializing the NR (or FDXB) solver and obtaining an accurate result for the PFP under these new conditions. Several simulations were carried out. The goal is always to minimize the number of points across the homotopy pathway because each point in the path demands

a LU factorization of the matrix $G_x(\mathbf{x}(t))$. The dynamical homotopy-based technique used the NR and FDXB solvers to refine the solution. Again, in the instant t_1 , the states $\mathbf{x}(t_1)$ were always obtained using only the BE scheme.

In the first simulation, the homotopy pathway was set with the points defined by the instants $t_k \{0, 0.05, 1.0\}$. A very large time-step, $\Delta t_1 = 0.95$, is verified for determining t_2 . Therefore, it is necessary to compute the states for the two points $t_1 = 0.05$ and $t_2 = 1.0$ to reach an approximate state $\hat{\mathbf{x}}^{(0)}$ at the latter instant. The simulations were performed with the solvers FE, RK2, and BE, using IC departing from $\mathbf{x}(t_1)$. For this scenario, only the BE scheme could provide an approximate result, which led to the successful convergence of the NR or the FDXB solvers.

In a second set of experiments, the homotopy pathway now was set with four points and defined by the instants t_k for three different forms: $\{0, 0.05, 0.1, 1.0\}$; $\{0, 0.05, 0.15, 1.0\}$; and $\{0, 0.05, 0.20, 1.0\}$. Again, among FE, RK2, and BE schemes, only the latter was able to provide an approximate result for the successful convergence of the NR or the FDXB solvers.

Another set of experiments investigated the homotopy pathway using five points. The values of t_1 and t_2 were held constant at 0.05 and 0.1, respectively, while only the value of t_3 was changed, with t_4 set at 1 as the final path point. The experiments considered the following time instants t_k for three different path options: $\{0, 0.05, 0.1, 0.20, 1.0\}$; $\{0, 0.05, 0.1, 0.25, 1.0\}$; and $\{0, 0.05, 0.1, 0.30, 1.0\}$. The simulations considered for these options allow for obtaining a suitable approximation of states $\hat{\mathbf{x}}^{(0)}$ to the NR or FDXB solvers. So, convergent solutions with high accuracy for all PFPs were calculated to a test system with explicit and implicit schemes. The results concerning the *case109k*, *case36964*, and *case_ACTIVSg2000limit* are discussed in the sequel. Figure 3 depicts plots indicating the evolution of the infinite norm of the PFP mismatches for the three test systems when the NR solver is initialized by an explicit (FE or RK2) or implicit (BE) integration scheme. The homotopy pathway has five points, but only the instant t_3 was modified along the simulations, being assigned the values 0.20 and resulting in Figure 3a; 0.25, for Figure 3b; and 0.30, for Figure 3c. In the plots, the norms are associated with results obtained through the dynamical homotopy-based methods and the NR scheme's iterations. Then, the term "Iterations" in the axis of abscissa means index k of a time instant t_k when the visualization occurs from 0 to 4 and means effective iteration of the NR solver for an index greater than 4. In the plot legends, the mnemonics "109k", "36k", and "lim" refer to identification for the *case109272*, *case36964*, and *case_ACTIVSg2000limit*, respectively. Still, from the plots, the norm results for the index 0 in the abscissa axis correspond to the infinite norm of the mismatches for the original estimate of the PFP, which is initialized with a flat start. The result for index 4 is associated with the instant $t_4 = 1$, the latest point of the homotopy pathway, and also to the approximate states $\hat{\mathbf{x}}^{(0)}$. This is then used as an initial estimate for the NR solver.

The plots in Figure 3 indicate that the performance of the dynamical homotopy-based schemes provided convergence for the NR solvers with a number of iterations closer to each system, except for the case where $t_3 = 0.25$ and the FE scheme was employed. Another remark is that the RK2 scheme for more elevated t_3 presented a performance similar to BE. Then, this similarity is expected to stay stronger by increasing the number of points for the homotopy pathway. However, this procedure must be avoided since the objective is to minimize the number of points of the homotopy pathway, aiming to reduce the computational cost in the global PFP resolution.

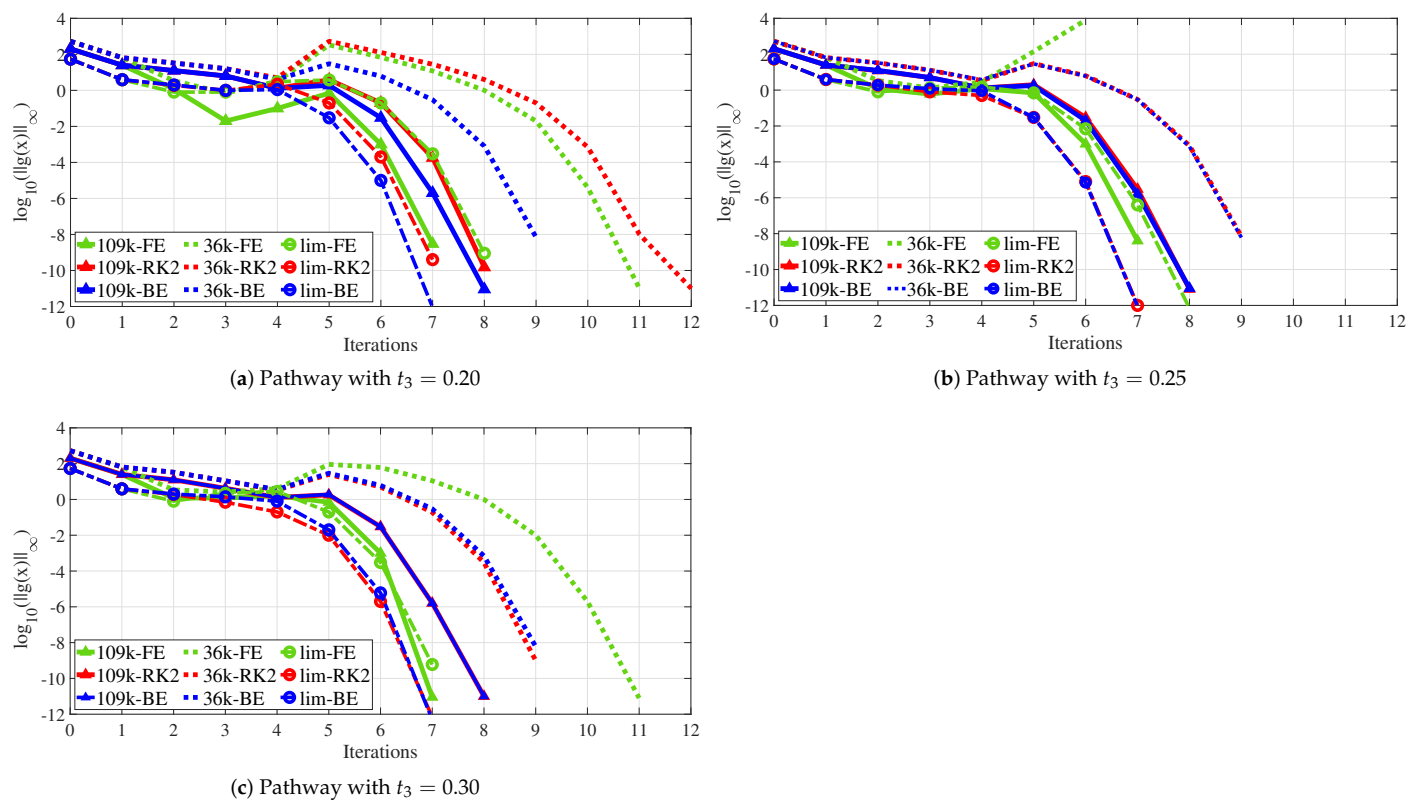


Figure 3. Plots indicating the evolution of the infinite norm of the PFP mismatches for three test systems when the NR solver is initialized by an explicit (FE or RK2) or implicit (BE) integration scheme, considering the homotopy pathway with five points, but only the instant t_3 is changed.

Table 6 shows the required global mean CPU time (dynamical homotopy-based plus NR scheme actions) to perform the simulations for obtaining the PFP solution according to the experiments used to generate the plots in Figure 3. Other experiments include simulations with the power flow executed using the native initial estimate provided by the MATPOWER data bank (NR-MAT), which this paper uses as a reference framework. The initial estimate in this condition is closer to the known solution. All other mean CPU times result from the computations in which the PFP starts using a flat start guess.

Table 6. Mean CPU time in seconds for running experiments whose results were presented in the plots of Figure 3.

CASES				
METHOD	t_3	case109272	case36964	case_ACTIVSg2000limit
NR-MAT	-	3.8889 (100%)	1.8862 (100%)	0.3100 (100%)
FE	0.20	5.6750 (146%)	3.3580 (178%)	0.2120 (68%)
	0.25	5.6500 (145%)	fail (-)	0.2340 (75%)
	0.30	6.6210 (170%)	3.3680 (179%)	0.2240 (72%)
RK2	0.20	8.9580 (230%)	4.3800 (232%)	0.2550 (82%)
	0.25	8.7950 (226%)	3.5900 (190%)	0.2610 (84%)
	0.30	9.2150 (237%)	3.6770 (195%)	0.2270 (73%)
BE	0.20	6.6550 (171%)	2.7160 (144%)	0.1890 (61%)
	0.25	6.6540 (171%)	2.7110 (144%)	0.2500 (81%)
	0.30	6.5690 (169%)	2.7860 (148%)	0.2900 (94%)

From Table 6 it is verified that the explicit integration method FE scheme has almost the same computational burden as BE. However, a comparison is made for the simulation with many points. It was shown that only the BE works properly for smaller points. Then, it is expected to have better BE performance than FE. The RK2 scheme requires more CPU time. This is because RK2 involves the computations of constants K_1 and K_2 , while the simplified scheme implemented for the BE is almost similar to FE. Similar to FE, RK2 does not work well for a small number of points, as happens in the BE scheme. Therefore, the results in the table confirm that the BE solver is suitable for use in the dynamic homotopy-based technique. Consequently, the results show that the CPU time can be even smaller when the BE scheme is used with fewer points in the homotopy pathway.

We can observe that from the results of the several sets of experiments performed with proportional increment for Δt_0 and K , only the BE scheme was able to present a suitable approximation to the NR solver for all situations. The schemes FE and RK2 had an adequate performance only for a larger number of points in the homotopy pathway. Yet, it required considerably small time steps for the initial time instants. This requirement was not essential when using the implicit BE method, except when considering the instant t_1 . It was verified that when many points of the homotopy pathway are used, all integration methods provide almost similar performance for searching a good initialization $\hat{\mathbf{x}}^{(0)}$. On the other hand, a large number of points must be used only in the case obtaining an adequate solution is infeasible; that is not the situation verified since the implicit scheme BE proved to be efficient for starting the NR and FDXB solvers even with three points (say, instants $\{0; t_1; 1.0\}$) in the homotopy pathway.

The simulations were also assessed by increasing the values of K to 0.002 and Δt_0 to 0.1 while keeping the perturbation δ at 0.02. However, the large value of Δt_0 did not contribute effectively to determining a suitable approximation of states $\hat{\mathbf{x}}^{(0)}$, which made it impossible to properly initialize the NR and FDXB schemes. When using the dynamical homotopy-based scheme for simulation, it is crucial to take into account not just the parameter δ , but also the initial time-step Δt_0 to obtain a suitable initialization $\hat{\mathbf{x}}^{(0)}$. Once this is achieved, the factor K can be calculated.

4.5. Simulations for a High Time-Step Dynamic Homotopy

Simulations were performed for all ill-conditioned test systems to check the impact of using a high-time step. Given the previous findings, the BE solver was utilized as the integration method. In the experiments, we employed a small time-step $\Delta t_0 = 0.005$, $K = 0.0001$, and a very large $\Delta t = 1 - \Delta t_0 = 0.995$ to determine the two points ($t_1 = 0.005$ and $t_2 = 1.0$) of the homotopy path curve $\gamma(t)$, $0 \leq t \leq 1$. The initial condition for the dynamic homotopy problem is given as $\mathbf{x}(0) = \mathbf{x}^{(0)}$, where $\mathbf{x}^{(0)}$ is considered as the flat start type for the PFP. The states obtained at time $t_2 = 1.0$ were used as an initial estimate for the NR solver to compute a high-precision solution for the PFP. To illustrate the evolution of specific states, the results for the largest system were combined and presented in plots, as shown in Figure 4. Power mismatches along the convergence process were also included in the same graphics space.

In the plots depicting the evolution of states, both the contribution of the dynamic homotopy and the NR refining actuation are distinguished by using a combination of dotted lines and solid lines. The dotted lines represent the dynamic homotopy, while the solid lines represent the NR solver contribution. Since it is a hybrid visualization process, including time instants and iterations on the same axis, it was agreed to display only the designation “iterations” on the abscissa axis. When using the dynamic homotopy process, the index represented by k as an iteration on the mentioned axis should be understood as the corresponding results for the time t_k . The ordinate axis on the left-hand side of the graph shows information about a specific state, while the mismatch norm is identified on the right-hand side, this latter in logarithmic scale.

Figure 4 exhibits the plots of five states and the power mismatches along with the iterations for the 109,272-bus system. The Figure 4a is the voltage magnitude, and Figure 4b

is the phase angle for the buses 2 to 6 (all of type PQ). The flat start nature of the original PFP guess is characterized by observing the voltage magnitudes starting at 1.0 pu at $t = 0$ with a phase angle of zero.

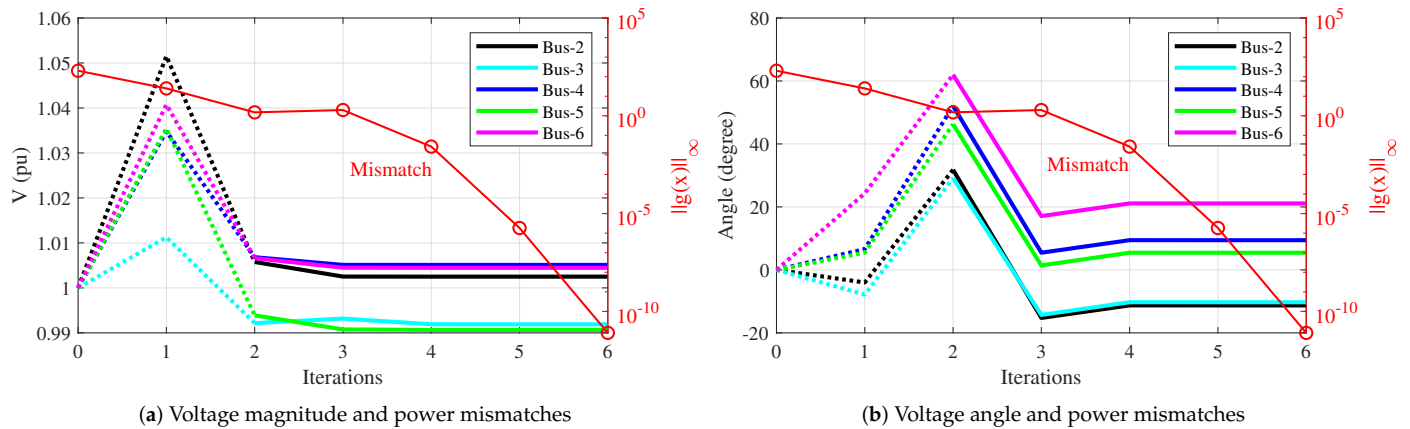


Figure 4. Results for the evolution of states and decaying power mismatches for the case109272, when the BE solver is used for the dynamic homotopy process and the results are refined by the NR solver.

It is possible to observe the evolution of the mismatch $\|g(x)\|_\infty$ as the states start from the initial flat voltage estimate and, at each iteration, converge to the solution of the problem. The dotted-line in the plot indicates the results for two time instants obtained by the BE solver. At time t_2 , corresponding to the second “iteration” (index 2) in the abscissa axis, we calculate voltage magnitudes with low accuracy. This leads to a power mismatch of about 1.0 (as shown by the red curve). The results are used as the initial estimate for the NR solver (solid lines), which only requires four iterations to reach convergence. The power mismatches decrease from 1.0 to less than the tolerance of 10^{-8} pu. Note that with relation to the voltage magnitude states, the dynamic homotopy process provides good accuracy for the NR solver obtaining convergent states (see solid lines in Figure 4a). However, it also successfully guides the phase angle states to convergent solutions despite large excursion on the variables at the point t_2 (see Figure 4b).

It has been confirmed that partial results are generated by calculating only two points while processing the dynamic homotopy. These results are sufficient to ensure that the Newton-Raphson solver converges quickly and effectively, demonstrating the effectiveness of the initial estimation for the NR solver.

Therefore, we can conclude that the dynamic homotopy process contributed to the successful convergence of the NR solver. Without this initialization, starting directly with a flat guess causes the NR solver to fail to converge for the 109,272-bus system.

4.6. Impact of Different Time Steps on the Improvement of Convergence

This subsection investigates the sensitivity of the time step with different sizes in the dynamic homotopy process. Experiments are carried out to show the BE solver’s performance on the states and the NR solver’s contribution in the final computations.

As before, the dynamic homotopy steps were used to find a good estimate for starting the NR solver. To do this, we performed experiments where the main time-step for $\Delta t > \Delta t_0$ was changed. For these experiments, Δt was kept constant for five different simulations, with time steps of 0.1, 0.125, 0.25, 0.5, and 0.995, assuming an initial time-step of Δt_0 . All the simulations were done with $K = 1 \times 10^{-4}$.

To follow the profile of a state along the pathway (instant of times) and the iterations of the NR solver, the evolution of a given state was monitored. The voltage magnitude and phase angle at the bus #6 of the 109,272-bus model were selected.

Figure 5 shows the plots for the chosen states and assigned Δt . As in Figure 4, in the figure's graph, the label "Iterations" on the abscissa axis refers to either an index k , representing results for a time instant t_k of the dynamic homotopy solver (shown with dotted lines), or to an iteration when the NR solver is invoked (shown with solid lines).

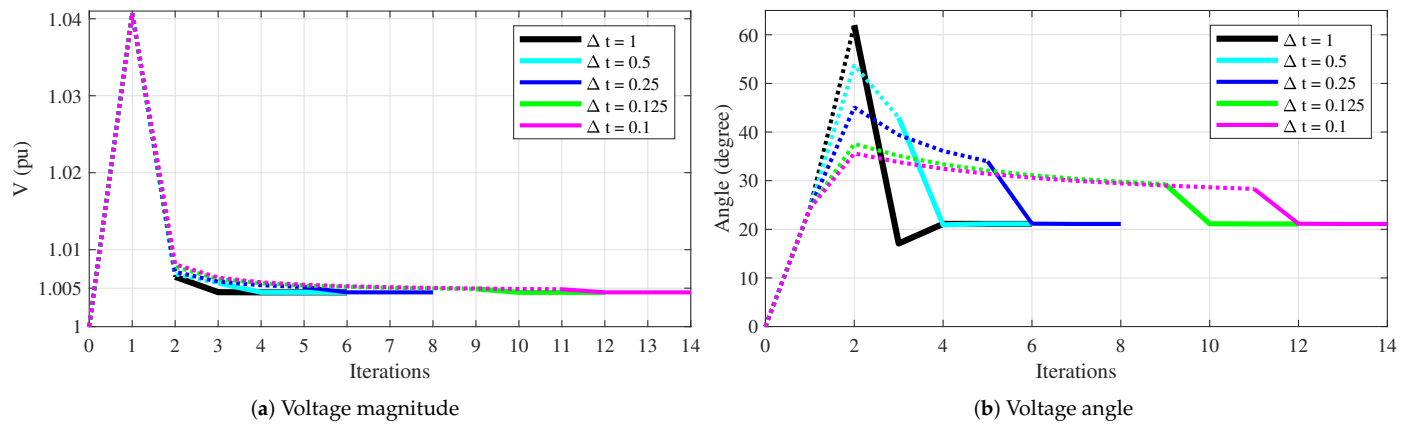


Figure 5. Evolution of the voltage magnitude and phase angle with the index k of a time instant t_k of the dynamic homotopy solver BE (dotted lines) or iterations for the NR solver (solid lines). Results are for the bus #6 of case109272, with different Δt .

Analyzing the plots of Figure 5, clearly all the curves converge for the same values for the states, about 1.005 pu for Figure 5a and approximately 20 degrees for Figure 5b. The critical condition can be attributed when $\Delta t = 1$ (black line), which provides to the NR method an initial estimate close to 1.04 pu and 60 degrees. Despite this, the NR solver confirms the final convergence close to 1.005 pu and 20 degrees. However, when increasing the number of homotopy steps, for example, using $\Delta t = 0.1$ (pink line), the BE method gets an initial estimate close to 1.005 pu and 28 degrees. Hence, near to the objective solution. On the other hand, more points t_k in the homotopy path are necessary. Even so, despite the diversified values for the Δt , the low-precision results provided by the dynamic homotopy approach were sufficient for the successful convergence of the NR solver.

Then, as the step size is reduced, the number of iterations in the dynamic homotopy process increases. In this case, the NR stage tends to begin its iterative process with a refined initial estimate, closer to the solution. However, as demonstrated, such a strategy does not bring any additional computational benefit since it demands more LU factorizations without altering the final results.

Therefore, there is evidence that the NR solver successfully utilizes the low-accuracy results of the BE solver to reach converged states. This is achieved by adjusting the time step, either making it large or small.

4.7. Experiments with Other Numerical Methods

To show how effective the partial result obtained by the dynamic homotopy method is for starting another iterative method, in this subsection, simulations with the classical NR method and its faster-decoupled version (FDXB, used in MATPOWER [30]) are assessed. Furthermore, other methods for comparison purposes were used.

The dynamic homotopy solver experiments yielded results that enable us to initialize other methods and achieve precise solutions that the classical NR method cannot attain alone. Simulations using BE and RK2 are performed. Due to the numerical limitations observed for the RK2 solver in the experiments, the BE and RK2 schemes were utilized with different time steps. Partial results, $\hat{x}^{(0)}$, were then obtained using these techniques to initialize the NR and FDXB solvers.

Again, the native initial estimate results provided by the MATPOWER (NR-MAT) were used as a reference. Other simulations include the NR solver starting directly from a flat start estimation (NR-flat), and a homotopy-based version called Guided Solution

Homotopy by using NR (GSH-NR) [12], which employs a root-finder for computing the precise roots of the homotopy pathway. Note that the dynamic homotopy solvers calculate approximations of the pathway roots since the computations process is carried out by numerical integration methods without the exigence of high accuracy in the results. Therefore, taking advantage of a simpler approach than the root-finder used in an NR-based solver properly, as GSH-NR.

Table 7 shows the results in terms of iterations for both convergent cases and those ones with fails. The 2nd column in the table indicates the number of iterations required for the NR solver (NR-MAT) to reach the convergence, assuming initial conditions as assigned in MATPOWER's databank. In the 3rd column is presented the result for the classical NR solver when a flat start (NR-flat) is assumed. From the 4th to 6th columns, the results are presented for the GSH-NR solver. In 4th column, the step for a homotopy pathway is given. The 5th column shows the parameter δ used in [12] to reduce impact of impedance connecting buses in the neighborhood of the slack bus. The 6th column exhibits the global number of iterations for the NR solver calculate all homotopy path points for the problem, which at the end produces the solution of the PFP. From 7th to 10th, the number of iterations necessary to refine the partial solutions found by BE and RK2 techniques ($iter_{ref}$) are shown, when NR or FDXB is used. All simulations were carried out just with $K = 0.0001$. In investigating the dynamic homotopy solvers, simulations were conducted at time steps of 0.005 and 1.0 for the BE solver (only two points in the homotopy pathway). The RK2 solver necessitates more time-steps since it requires small time steps at the initial transient phase. Consequently, time steps of 0.005, 0.01, 0.02, 0.12, 0.42, 0.72, and 1.0 (seven points in the homotopy pathway) were considered for the RK2 solver. Simulations using FE were not presented due to their results being almost similar to those of RK2, albeit with reduced performance.

Table 7. Number of iterations ($iter_{ref}$) for the refining solvers (NR or FDXB) starting from partial results obtained by a dynamic homotopy solver for the experiments involving ill-conditioned and overloaded systems. Additionally, the iterations resulting from convergent actions of NR-MAT, NR-flat, and GSH-NR are shown.

CASE	NR-MAT	NR-Flat	GSH-NR		BE _(NR)	RK2 _(NR)	BE _(FDXB)	RK2 _(FDXB)
	$iter_{NR}$	$iter_{NR}$	Δh_1	δ	$iter_{NR}$	$iter_{ref}$	$iter_{ref}$	$iter_{ref}$
case18482	6	fail	0.5	1	10	4	3	33
case27318	5	fail	0.25	0.125	25	4	3	29
case36964	6	fail	0.25	1	27	4	3	33
case54636	5	fail	0.01	0.01	fail	4	3	29
case109272	5	fail	0.01	0.01	fail	4	3	19
case69limit	4	4	1	1	4	1	1	8
case141limit	3	3	1	1	3	1	1	8
ACTIVSg500limit	3	4	1	1	4	2	2	13
ACTIVSg2000limit	3	5	1	1	5	3	3	22

Based on Table 7, we can see that convergence fails only in the simulations using the NR-flat solver, which comprises the cases with the NR solver starting directly from a flat estimation. Convergence also fails for the cases associated with the largest models when using the GSH-NR solver. Despite trying different parameters such as Δh_i and δ , convergence was unsuccessful in this latter situation. On the other hand, all other simulations using the NR-MAT, and the dynamic homotopy NR (or FDXB)-based solver, achieved successful convergence.

The information in Table 7 demonstrates that using the partial results of the dynamic homotopy BE solver, the NR solver required only four iterations (see 7th column) to refine the low-accuracy solution. It is emphasized that BE was implemented with only two points ($t_1 = 0.005$ and $t_2 = 1$) in the homotopy pathway to calculate the partial solution. Furthermore, when we analyze the performance of the FDXB solver using the same BE,

the high number of iterations found (see 9th column) apparently conveys the notion that the method has poor performance. On the other hand, as it will be observed ahead of the measurement of the CPU time required for the calculations in the upcoming experiments, the method demonstrates exceptional performance. It achieves the same expected result for the PFP with the same precision as the classical NR solver and offers a simplified scheme, leading to reduced computational costs. It is important to note that the combined solver $BE_{(NR)}$ starts from a flat estimation. Despite this, it still requires the same number of iterations (4 iterations for refinement plus 2 evaluations of BE) as the NR-MAT, even though the NR-MAT starts with an estimate close to the solution.

When analyzing the RK2 solver's capability to generate good partial results for initializing either NR or FDXB, the iterations in the 8th and 10th columns of Table 7 are observed. It is worth noting that the RK2 alternative also requires a similar number of iterations as BE. Precisely, when using the NR solver option, only 3 iterations are needed for RK2, while 4 iterations are required for the BE option. However, the dynamic homotopy solver requires 7 points for RK2, while only 2 for BE. Additionally, RK2 needs more calculations to obtain the result at a given pathway point than BE. When dealing with FDXB, the number of iterations is quite similar when starting with BE. Then, considering the computational load related to the dynamic homotopy, the $BE_{(FDXB)}$ option is more advantageous than $RK2_{(FDXB)}$.

4.8. Computational Burden Analysis

This subsection focuses on experiments dedicated to measuring the mean CPU time. The main purpose is to analyze the performance of the combined methods for computing a high-precision PFP solution. The solvers used in the calculations are the same as those used to generate the results in Table 7. However, the ongoing simulations were evaluated under different conditions to measure the CPU time because the measurements were obtained considering the mean of a hundred repetitions of the same simulation.

The original PFP MATPOWER calculation starts with an initial estimate assigned in the neighborhood of the solution using the NR-MAT solver. Then, it was used as a reference result for CPU time measurements. In contrast, the other results were obtained from a flat start estimate, which does not allow for convergence for the NR method without the transient-based approximated states for the five test systems and, finally, the GSH-NR scheme.

The average CPU time is displayed in Table 8. Each pair of columns in the table, starting from the second one, shows the CPU time in seconds and the corresponding percentage, relative to the load of the NR-MAT (the reference CPU time solver).

Table 8. Mean CPU Time in seconds for running experiments considering ill-conditioned test systems and solvers.

METHOD	CASES									
	case109272		case54636		case36964		case27318		case18482	
NR-MAT	3.8889	(100%)	2.1236	(100%)	1.8862	(100%)	0.8912	(100%)	0.8241	(100%)
GSH-NR	fail	(-)	fail	(-)	40.6900	(2157%)	21.5540	(2418%)	9.4790	(1150%)
$BE_{(NR)}$	4.2730	(110%)	2.1840	(103%)	1.7260	(92%)	1.1430	(128%)	0.9040	(110%)
$RK2_{(NR)}$	6.3920	(164%)	3.1070	(146%)	4.1100	(218%)	1.6240	(182%)	1.1630	(141%)
$BE_{(FDXB)}$	2.6890	(69%)	1.4070	(66%)	1.0840	(57%)	0.7560	(85%)	0.6250	(76%)
$RK2_{(FDXB)}$	4.1690	(107%)	2.0840	(98%)	1.5890	(84%)	1.0600	(119%)	0.9160	(111%)

With initialization given in MATPOWER (NR-MAT), the NR solver demands 3.89 s (100%) to calculate the PFP solution for the 109,272-bus system (see 3rd line). The GSH-NR fails for this resolution. However, the composed NR solver using the dynamic homotopy solution with BE plus the NR solver (see 5th line) has a computational cost of 110%, almost similar to the reference solver. When the homotopy-based solver is changed to RK2, the computational increases to 164%, demonstrating the better performance of the BE in relation

to RK2. This fact is justified because the former demands less LU factorization per calculated point. In this work, only one LU factorization is employed, despite the BE solver being an implicit method. However, the surprising result is attributed to the solver that uses the FDXB method with the dynamic homotopy technique as the starting point. In combination with the BE solver, the $BE_{(FDXB)}$ takes only 69% of the reference. The $RK2_{(FDXB)}$ solver is not as effective because it requires more computations in the dynamic homotopy process, using more steps than the BE solver. In all cases, a better computational performance than the reference method is achieved by the $BE_{(FDXB)}$ scheme, requiring the lowest cost for case36964, 59%; and the highest cost for case27318, 85%. For the convergent cases, the computational burden for computing the PFP solution was the highest when using the GSH-NR solver in the investigated cases.

In the cases studied evaluating the combination of the number of iterations plus the dynamic homotopy activity, the RK2 always performed worse than BE. This was also confirmed when the comparison was the CPU time. However, combined with FDXB, note that a better performance than the reference solver was achieved for case36964, with 84%; and case54636, with 98%.

Based on the evaluations, using dynamic homotopy in the initialization process for the NR and FDXB methods has proven highly effective. Furthermore, the approach allows a simplified method like FDXB to achieve convergence with the same precision as NR while demanding less computational effort than NR-MAT.

5. Conclusions

This paper presented a dynamic homotopy-based technique to improve the convergence for the classical Newton-Raphson solver. The improvement obtained is also adequate, even for simplified methods derived from the full NR scheme as the FDXB solver. The scheme differs from the usual homotopy problem, where an algebraic equation is solved to compute the pathway points precisely. In the dynamic homotopy, the points are approximations, and at the end of the path, the states are used as initial estimates for the NR (or FDXB) method.

Explicit and implicit integration methods were investigated to solve the dynamic homotopy problem. This problem is formulated with the ODE IC coinciding with the traditional PFP initial estimate. The studies assumed this estimate to be a flat start, which can be considered a hard situation compared with the starting conditions used originally in MATPOWER's databank or the original cases presented in this paper. The explicit integration methods used were the forward Euler (FE), the second-order Runge-Kutta (RK2), and the implicit backward Euler (BE). Considering simulations for ill-conditioned and large-scale systems, besides systems for stressed scenarios, the BE solver was the scheme that demonstrated the best performance for the test systems. Initializing the full NR and the simplified FDXB solvers proved effective. Besides presenting excellent qualities of convergence, this latter performed better than the former when the mean CPU time was measured.

In future works, the authors plan to investigate other types of functions for implementing the dynamic homotopy-based problem and other integration methods to solve the dynamic homotopy scheme. Furthermore, our goal is to apply the technique to larger networks.

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